

# TFPT — Research Contracts for the Remaining Interfaces:

$$v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$$

( $U_{\text{wall}}$ ): the parabolic flavor wall-selection    and    ( $G_{\text{metric}}$ ): the full quantum-gravity measure

Stefan Hamann

Alessandro Rizzo

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (**verification/status\_ledger.csv**). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading companion R (Research Contracts).**

### TFPT status at a glance — read this card the same way in every document

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ( $[E]$ ); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status          |
|-----------------------|-------------------------------------------------------------------------------------------------|-----------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive $[O]$ |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | $[E]/[C]$       |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | $[C]$           |

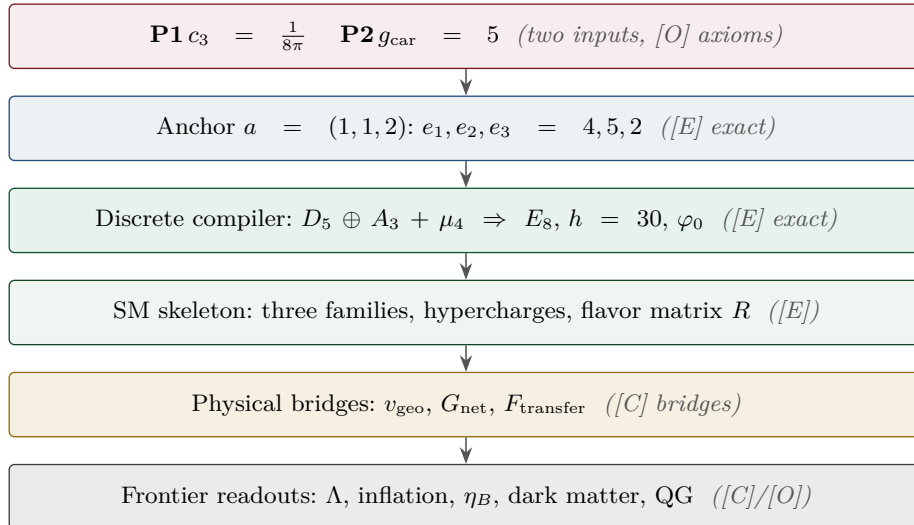
Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}$ ,  $G_{\text{metric}}$ ,  $F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ .  
 Marker key:  $[E]$  exact / machine-proven ·  $[C]$  conditional (holds under named hypotheses) ·  $[O]$  open / axiom ·  $[X]$  falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

$[E]$  exact / machine-proven

$[C]$  conditional

$[O]$  open / axiom

$[X]$  kill test



### What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. ( $F_{\text{frontier}}$ ) (Koide,  $\eta_B$ , axion relic scale,  $m_p/m_e$ ) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** ( $U_{\text{wall}}$ ) *first* (finite, algebraic, falsifiable), then ( $G_{\text{metric}}$ ) (deep analytic programme).

**Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity).** The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

—  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ : the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71),  $U_{\text{point}}$  collapses to the *one* dimensionful unit  $v_{\text{geo}}$  shared with  $1/G$  (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where  $U_{\text{point}}$  still appears);  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ : holographically reduced to the single boundary-net statement “*the seam-Calderón inclusion has Jones index 4 =  $|\mu_4|$* ” — holomorphy,  $(E_8)_1$  and the unique 2D bulk then follow (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier  $\rightarrow SO(16)_1 \rightarrow (E_8)_1$  carries one and the same 16-fermion content, whose single quasi-free kernel (rank =  $c$ ; Wick/Pfaffian exact) is the certified Calderón datum — closing  $G_{\text{net}}$  therefore also closes the carrier choice [E] ([verification/v113\_quasifree\_kernel.py]);  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ : named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the  $K \rightarrow C \rightarrow F$  sheet axis  $M(1, t)$ , with the Koide source  $\rightarrow$  pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([verification/v224\_diamond\_ftransfer\_path.py]).

**QFT layer (this round).** The boundary QFT is now *one relative object*  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ , closed modulo the *single* keystone **SEAM.EQUIV.01** (whose downstream conformal-deck face is **QGeo.SYM.01**): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and  $E_8$  live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure  $(G_{\text{metric}})$  is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

## Research Contract 1 — $(U_{\text{wall}})$

**Goal.** Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{ cusp}, \det R=8, \text{ Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$  and  $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$  at the four points of  $\mu_4$ . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta-Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one*  $D_4$ -symmetric, rank-3, four-point point that realises the selectors.

**Lemma 1** (U0 — normalisation). *The gate data are equivalent to the  $SU(3)$  character variety  $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$  with  $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$ ,  $\omega = e^{2\pi i/3}$ , and  $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$ . To show:  $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$  as polystable Mehta-Seshadri / tame Hodge realisation. Machine: the group presentation and  $D_4$ -action are finite and codable (Sage/Wolfram).*

**Finite rigidity — now proven [E].** The finite half of U0 is closed exactly:  $\mu_4$  has cross-ratio 2 and Möbius stabiliser  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of order  $8 = |D_4|$  (a 4-cycle of the marks + a reflection, faithful);  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$ ,  $k = 1, 2, 3$ , are  $\mu_4$ -eigenforms ( $z \mapsto iz : \omega_k \mapsto i^k \omega_k$ ) carrying the three nontrivial characters  $\chi_1, \chi_2, \chi_3$  with weights  $(1, 2, 3) = \text{Spec}(Q_+)$ . Hence a genus-zero boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  with the  $(1, 2, 3)$  grading (**QGeo.RIGID.01**, [E], [verification/v168\_qgeo\_rigidity.py]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays **QGeo.REALIZE.01** ([C]).

**Seam Collar Realisation Theorem — the one central proof obligation [E]/[O]**  
([verification/v176\_seam\_collar\_realisation.py])

*Stated as one theorem and certified as a reduction — not a fabricated proof.*

**Theorem (target).** *Given* (1) an oriented one-sided RP seam collar  $\Sigma$ ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap  $(2/3)^6$ ; (4) a faithful  $D_4$  action on the boundary marks; (5) four parabolic marked points with  $\mu_4$ -character decomposition; (6) an admissible  $c=8$  boundary net — *then* the conformal boundary of  $\Sigma$  is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$ , its  $H^1$  character basis is canonically the generation basis with grading  $(1, 2, 3)$ , the seam Calderón contraction is the

rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$ .

**The proof decomposes into six steps, five already [E]:**

- 1–2 *Geometry / uniformisation* [E] (v168):  $\mu_4$  has cross-ratio 2 and a faithful Möbius  $D_4$  stabiliser of order 8; four genus-0 marks with faithful  $D_4$  force the  $\mu_4$  square up to Möbius;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ .
- 3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol  $|k|$  (harmonic extension  $e^{ikx - |k|y}$ ), the free chiral dispersion — bulk-detail-independent.
- 4  *$H^1$  character = generation grading* [E] (v137/v168):  $\omega_k = z^{k-1}dz/(z^4 - 1)$  carry  $\mu_4$  characters  $i^k$  with weights  $(1, 2, 3)$  = the  $A_3$  exponents =  $\text{Spec}(Q_+)$  — natural, not merely isomorphic.
- 5 *One-particle contraction* [E] (v162): a  $\mu_4$ -equivariant operator is diagonal in the cusp basis  $(\frac{1}{|\mu_4|} \sum_j U^j X U^{-j} = \text{diag } X \text{ for } U = \text{diag}(1, i, -1, -i))$ ; its spectrum is the cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , so the transfer eigenvalues  $(1 - \alpha)^{p_2}$  ( $p_2=6$ ) are  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $(2/3)^6$  — forced, no free knob.
- 6 *AQFT closure* [E] (v154/v175):  $(D_5)_1 \otimes (A_3)_1$  with the isotropic  $\mu_4$  glue has index  $4 = |\mu_4|$ ,  $c = 5+3 = 8$ ,  $\mu$ -index  $16/16 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  ( $248 = 120+128$ ); full-cone RP holds for all  $m$  via the CAR second-quantisation functor.

**The proof tree — and the two genuine open doors [O]** ([verification/v177\_seam\_marking\_kernel.py]). Taking the marks/ $D_4$  as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\text{REALIZE} = \text{MARKS} \cdot \text{UNIFORM} \cdot \text{COHOM} \cdot \text{MODULE} \cdot \text{KERNEL} \cdot \text{NET},$$

of which *four nodes are now closed constructively*: UNIFORM [E] (an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma: z \mapsto 1/z$  gives a faithful  $D_4$  — so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}^1, \mu_4)$ , strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as FORM.QGEO.02:  $\rho^4 = \text{id}$ ,  $\rho^2 \neq \text{id}$ ,  $\sigma^2 = \text{id}$ ,  $\sigma\rho\sigma = \rho^{-1}$ , the orbit scaling to  $\mu_4$ , the  $\mu_4$  permutation, and the multiplier classification “order-4  $\Leftrightarrow \zeta = \pm i$ ” are machine-proved in TftptCarrier/MobiusUniformisation.lean, AUDIT: PASS); COHOM [E] ( $\rho^* \omega_k = i^k \omega_k$ , grading  $(1, 2, 3)$ ) — also **Lean-formalised** as FORM.QGEO.03: the pullback characters  $i^1, i^2, i^3$  and the reflection parity  $\sigma^* \omega_1 = \omega_3$ ,  $\sigma^* \omega_2 = \omega_2$ ,  $\sigma^* \omega_3 = \omega_1$  are machine-proved in TftptCarrier/CohomologyGrading.lean, AUDIT: PASS); MODULE [E] (a *unique*  $\mu_4$ -equivariant iso — multiplicity-free + residue normalisation, reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed; the parity is the Lean FORM.QGEO.03 above, the *uniqueness* stays symbolic); and NET [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: QGEO.MARKS.01 — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked  $D_4$  boundary (the four marks from the anchor atom  $e_1(a)=4=|\mu_4|$ , collisions excluded,  $\tau\rho\tau = \rho^{-1}$ ) — and QGEO.KERNEL.01 — the *raw* seam Calderón operator equals the  $\mu_4$ -equivariant free  $c=8$  gapped one-particle contraction *as an operator* ( $C_\Sigma = U^{-1}C_{\mu_4}U$ ), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

**How far each door opens by computation** ([verification/v178\_marks\_kernel\_attempt.py]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For MARKS*: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (not the fixed points), distinct iff  $b_1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving faithful  $D_4$  — so MARKS reduces to the *single* topological premise “the seam double is genus-0 with an order-4 clock whose marks form one orbit”. *For KERNEL*: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the “spectrum vs operator” distinction *vanishes* on the finite block. So KERNEL reduces to the *single* premise “the full DtN on  $L^2$  reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation” (principal symbol  $|k|$ , Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

**Marks without a geometric posit: a Lefschetz/character argument** ([verification/v195\_marks\_lefschetz\_character.py]). The marks need not be *assumed* as  $D_4$  boundary points: they follow from the  $H^1$  character alone. The order-4 clock  $\rho: z \mapsto iz$  fixes exactly  $\{0, \infty\}$ , so a puncture at a fixed point would contribute a *trivial* character to  $H^1$ ; but the carrier

datum is  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents  $(1, 2, 3)$ , with  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ), which has *no* trivial component — so no puncture sits at a fixed point. With  $\text{rank } H^1 = n - 1 = 3 \Rightarrow n = 4$  punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free  $\mu_4$  orbit (cross-ratio 2), and the closed-surface Lefschetz number  $L(\rho) = 1 + 1 = 2 = |\{0, \infty\}|$  is consistent. So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder *representation* premise “the raw seam  $H^1$  carries the  $A_3$ -exponent rep” — a carrier-algebra statement, strictly less circular, still [O].

**And the two doors are in fact ONE** ([verification/v179\_conformal\_realisation.py]). MARKS’s residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$ , and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the *same*  $\chi = 2$  that supplies the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points  $\{0, \infty\}$ , and a free order-4 orbit has exactly  $4 = e_1(a)$  points. So MARKS reduces to “the carrier clock acts *conformally* on the genus-0 double”, and KERNEL then *follows* from that same geometry (Lee–Uhlmann  $|k|$  + the  $c = 8$  rigidity v158 + the cusped-surface spectral split, residual =  $H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). **Hence the entire structural residual of the theory is a single geometric premise QGEO.CONF.01:** *the raw RP seam normal double is conformally* ( $\mathbb{P}^1, \mu_4$ ) *with the carrier clock as the order-4 Möbius automorphism*  $z \mapsto iz$ . Given it, everything — genus-0, the four  $D_4$  marks, the  $(1, 2, 3)$  grading, the DtN block, the rank-8 polarisation and the  $(E_8)_1$  net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

**And even that conformal premise reduces to a milder isometry premise** ([verification/v180\_clock\_is\_mobius.py]). QGEO.CONF.01 need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is*  $\mathbb{P}^1$  — the target is *produced*, not assumed — and by Kerékjártó (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ . An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of  $\mathbb{P}^1$ , and an order-4 Möbius map is  $z \mapsto iz$  (elliptic, multiplier  $i$ , free orbit  $\mu_4$ ). So QGEO.CONF.01 is *implied* by the strictly milder, non-circular premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does *not* name  $\mathbb{P}^1/\mu_4$  and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory;** its surface realisation from the raw seam stays honestly [O].

**And one step further reaches bedrock** ([verification/v181\_clock\_is\_conformal\_symmetry.py]). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), i.e. the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification QGEO.SYM.01: *the seam’s conformal structure is the  $\mu_4$ -deck structure of the carrier clock* (the carrier  $\mu_4$  glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam’s conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain  $\text{REALIZE} \rightarrow \text{theorem} \rightarrow \text{MARKS} + \text{KERNEL} \rightarrow \text{CONF} \rightarrow \text{ISO} \rightarrow \text{SYM}$  (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

**A sharper, operator-checkable restatement:** QGEO.ENERGY.01 ([verification/v192\_qgeo\_energy\_clock.py]). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock  $\rho$  preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double,  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$* . In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality  $\rightarrow$  Möbius  $\rightarrow z \mapsto iz \rightarrow \mu_4$ ). Its appeal is checkability:  $\Lambda_\Sigma = |k|$  is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** “ $\rho$  preserves  $\Lambda_\Sigma$ ” is plausibly *equivalent* to “ $\rho$  is the conformal deck” (QGEO.SYM.01), so QGEO.ENERGY.01 *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if  $\rho$  is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays [O]; the gain is a sharper



operator-analytic target, not a closure.

**The non-circular sharpening: QGEO.ENERGY.02** ([[verification/v193\\_qgeo\\_energy\\_commutator.py](#)]). The way to make “ $\rho$  defined independently” precise is the *energy commutator*  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims: (1)  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  (order  $4 = |\mu_4|$ , v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* —  $\Lambda_\Sigma$  must be the DtN of the *raw* RP seam (definable from the RP data alone), not the  $\mu_4$  normal-form operator; (3)  $[\rho, \Lambda_\Sigma] = 0$  forces the  $\mu_4$  character grading  $(1, 2, 3)$  on  $H^1$ , already established (QGEO.COHO.01). An exact consistency rider pins the direction: the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (the *family* glue norm), while  $q(D_5) = \frac{5}{4}$  (the carrier norm) misses by  $g_{\text{car}}/N_{\text{fam}} = \frac{5}{3}$  — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds,  $[\rho, \Lambda_\Sigma] = 0$  turns QGEO.SYM.01 from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, [O].

**Subclaim (2) tackled — one notch closer, not closed** ([[verification/v194\\_raw\\_seam\\_dtn\\_rp.py](#)]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator  $T = e^{-H}$  from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression PTP, a contraction). So  $\Lambda_\Sigma$  is RP-definable without ever naming  $\mathbb{P}^1 \setminus \mu_4$ . The commutator  $[\rho, \Lambda_\Sigma] = 0$  then *closes on the finite cohomology*  $H^1$  [E]:  $\rho: z \mapsto iz$  acts on  $\langle w_1, w_2, w_3 \rangle$  ( $w_k = z^{k-1} dz / (z^4 - 1)$ ) as  $\rho^* w_k = i^k w_k$ , distinct characters  $i, -1, -i$ , so any  $\rho$ -equivariant operator is diagonal there (v177). What does *not* close is the *full- $L^2$*  commutation beyond  $H^1$  — it is exactly the statement that the quasi-free seam state’s modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to “intrinsic BW geometric modular covariance of the quasi-free seam state” — a named, standard (hard) AQFT property, one notch sharper, but still [O] (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

**A concrete variational handle on the residual** ([[verification/v196\\_seam\\_energy\\_functional.py](#)]). The open BW statement can be made a discretisable extremum problem via the failure functional  $E_{\text{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , whose zero locus is exactly the QGEO.SYM.01 conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite  $H^1$  block  $E_{\text{fail}} = 0$  at the  $\mu_4$  configuration ( $\rho = \text{diag}(i, -1, -i)$ ,  $\Lambda$   $\mu_4$ -equivariant,  $\Theta = \text{conjugation}$ ) and  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a genuine minimiser [E]; minimising  $E_{\text{fail}}$  over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

**The decisive crack: principal symbol exact + Tomita–Takesaki** ([[verification/v198\\_modular\\_commutator\\_reduction.py](#)]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is  $|k| = \sqrt{-\partial_\theta^2}$  (Lee–Uhlmann, v156); in the Fourier basis  $e^{in\theta}$  it is  $\text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz$  (rotation by  $\pi/2$ ) is  $\text{diag}(i^n)$ , both *diagonal*, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator  $\Delta_\omega$  is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\Delta$  and hence with  $\log \Delta \sim \Lambda_\Sigma$ : thus  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$  — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite  $H^1$  block is already  $\mu_4$ -invariant (v177), the entire residual is now the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega$ , equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of QGEO.SYM.01. A foundational symmetry postulate cannot be derived from nothing (the role  $c = \text{const}$  plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

**The residual, fully localised and numerically probed** ([[verification/v199\\_seam\\_state\\_invariance.py](#)], [[verification/v200\\_seam\\_variational\\_scan.py](#)]). Attacking  $\omega \circ \rho = \omega$  on the *raw* seam (not the  $\mu_4$  normal form) confirms it is non-circular —  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ ), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , to  $[\rho, H_1] = 0$ , i.e.  $H_1$  *is block-diagonal in the four  $\mu_4$ -character classes*  $\{n \equiv r \pmod{4}\}$ . The principal symbol  $|k| = \text{diag}(|n|)$  passes automatically, so the entire residual is the single bounded statement “the sub-principal symbol of the raw DtN has no off-character matrix elements”. The companion variational scan minimises  $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 + (\text{faithfulness})$  over a free clock angle and finds its *unique* minimum at  $\theta = \pi/2$  — the  $\mu_4$  clock  $z \mapsto iz$  (the only faithful order-4 grading  $i, -1, -i$ ) — a numerical sign that the variational

principle selects  $\mu_4$ . This is the “next executable theorem” a working mathematician can take up; it does not close the postulate, [O].

**The bounded sub-principal residual is the image of the  $\mu_4$  marks, not a free postulate** ([verification/v201\_seam\_subprincipal\_marks.py]). One genuine step further: the seam DtN is  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature function  $f(\theta)$  on the seam circle (the standard Steklov/DtN structure), so  $\langle n|M_f|n' \rangle = f_{n-n'}$ . Then  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  has Fourier support only on modes  $\equiv 0 \pmod{4}$ , i.e. iff  $f$  is  $\mathbb{Z}_4$ -invariant (the principal  $|k|$  already commutes, v198). But a curvature sourced by the four marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  for any profile  $g$ , has  $f_m = 4g_m$  [ $m \equiv 0 \pmod{4}$ ] — automatically  $\mathbb{Z}_4$ -invariant. So the  $\mu_4$ -mark orbit (forced by v195) forces the sub-principal symbol block-diagonal; 3 equally-spaced marks ( $\mathbb{Z}_3$ ) or 4 generic marks both break it (negative controls). The residual of  $\omega \circ \rho = \omega$  thus collapses to “the DtN sub-principal symbol is *mark-local* (the seam is flat away from the  $\mu_4$  marks = the conformal-deck structure, v177)” — a locality/flatness property, structurally definitional, the narrowest form yet of QGEO.SYM.01, [O].

**Certified on realistic Steklov profiles, and at the state level** ([verification/v210\_mark\_local\_dtn.py]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump  $g(\theta)$  localised at a puncture, summed over the four marks  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , has every off-(mod 4) Fourier coefficient vanishing to  $< 10^{-16}$  (exact 4-fold cancellation), the assembled DtN  $\Lambda = |D_\theta| + M_f$  has  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the *quasi-free state* it defines, with covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$ , is clock-invariant:  $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ , i.e. the actual  $\omega \circ \rho = \omega$ , not merely  $[\rho, \Lambda] = 0$ . A single off-centre bump, a  $\mathbb{Z}_3$  (three-mark) source and four unequally-spaced marks each break both by  $O(1)$ , and the mark-local commutator stays  $< 10^{-10}$  as the truncation grows ( $N \in \{16, 32, 64\}$ ) — not an artefact. This *numerically certifies the implication* “mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ ” for real profiles; the premise that the physical seam is mark-local stays [O] (it does not close QGEO.SYM.01).

**The same residual, read dynamically: modular flow + a recovery Lindblad generator** ([verification/v238\_modular\_lindblad\_dynamics.py]). The whole QGEO.SYM.01 discussion above is *static* — a commutator condition  $[\rho, \Lambda_\Sigma] = 0$ . Read as a *dynamics* it is the statement that the modular flow  $\sigma_t = \Delta^{it}$  (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian  $\Lambda_\Sigma = \log \Delta_\omega$ ) conserves the  $\mu_4$  clock:  $\rho \sigma_t \rho^{-1} = \sigma_t \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$  — so the open premise is exactly “modular time conserves the carrier charge”. The dissipative half is the recovery channel  $T$  of v221 read as a generator:  $L = \log T$  is a doubly-stochastic (CPTP-classical) Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time), and its dissipative gap is  $-\log((2/3)^6) = 6 \log(3/2) = \Delta$ , the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together  $G = i\Lambda_\Sigma + L$  is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real  $\leq 0$ ) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that  $\sigma_t$  is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra — is the same QGEO.SYM.01 bedrock, [O].

**From the dynamics to a QFT skeleton on the seam** ([verification/v239\_kms\_thermal\_time.py], [verification/v240\_gns\_os\_reconstruction.py], [verification/v241\_dhr\_sectors.py], [verification/v242\_spin\_statistics\_modular.py], [verification/v243\_haag\_ruelle\_braiding.py], [verification/v244\_spectral\_action\_contract.py]). With the static  $\rightarrow$  dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at  $\beta = 1$ , and the seam unit  $2\pi = 1/(4c_3)$  fixes the modular temperature as the horizon temperature  $T_H = c_3/M$ . *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from  $(\mathcal{M}, \omega)$ , and the OS Hamiltonian  $H_{\text{OS}} = -\log T = -L$  is positive with gap  $\Delta$  — the v64 “ $H \geq 0$  + mass gap” is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (FORM.GLUE.01); fusion is recovered from the modular  $S$  by Verlinde, the Gauss–Milgram sum returns  $c = 8$ , and the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (v235/v237) selects the holomorphic  $(E_8)_1$ . *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed  $(E_8)_1$ . The one genuinely 4d step is typed as a contract, CONTRACT.QFT4D.01 (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open [O] target. Its representation-theory core is now discharged ([verification/v245\_spectral\_matter\_content.py]): the **16** is exactly one *anomaly-free* SM generation ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ) with the forced SM hypercharges and the spectral-action value  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), so only

the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (QGeo.SYM.01) plus *one contract* (CONTRACT.QFT4D.01), not a diffuse list.

**The perturbative 4d S-matrix is constructible** ([[verification/v269\\_spert\\_paqft\\_skeleton.py](#)]). The 4d scattering layer is typed precisely as three *distinct* objects:  $S_{\text{top}}$  (the 2d DHR braiding monodromy,  $|M|=1$  statistics, v243),  $S_{\text{pert}}$  (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and  $S_{\text{phys}}$  (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (every operator mass dimension  $\leq 4$ : 7 marginal + 3 relevant, a *finite* counterterm basis), so [C] by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix  $S(g) = T \exp(i \int g L_{\text{int}})$  is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit. [O] this is *perturbative* only: the nonperturbative ambient measure QG.AMB.01 stays open, and the canonical 4d reading remains boundary-only (v265);  $S_{\text{pert}}$  is the perturbative cross-section layer on top (QFT4D.SPERT.01).

**A concrete one-loop EG number** ([[verification/v271\\_eg\\_oneloop\\_quartic.py](#)]). The skeleton is taken from “exists” to an actual Epstein–Glaser renormalization: the order-1 product  $T_1 = :L_{\text{int}}:$  needs no extension, and the first one is the order-2  $T_2$ , whose UV behaviour is the Steinmann *scaling degree*. [E] the massless propagator has  $\text{sd} = 2$  in  $d = 4$ , the one-loop bubble (two propagators) has  $\text{sd} = 4 = d$ , so the singular order  $\omega = \text{sd} - d = 0 \Rightarrow$  *exactly one* logarithmic local counterterm ( $C(\omega+d, d) = 1$  — finite, renormalizable, no infinite tower); the one-loop factor is  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly, the *same*  $1/(16\pi^2)$  that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop running is then  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry factor 3 = the  $s, t, u$  channels), with the EG initial condition fixed by the spectral action,  $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$ . [O] this is order-2/one-loop, one diagram class; the all-order construction and the nonperturbative measure stay open (QFT4D.SPERT.02).

*Red-team caveat* ([[verification/redteam/rt\\_F\\_qft4d.py](#)]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the  $R^2/\text{Weyl}^2$  terms give a four-derivative propagator  $1/(p^2(p^2+M^2))$  whose massive spin-2 mode has a negative residue (the Stelle ghost). So  $S_{\text{pert}}$  is a unitary perturbative S-matrix for the *matter+gauge* sector only; the gravity perturbative S-matrix carries the ghost — which is precisely why the *nonperturbative* QG.AMB.01 (asymptotic safety v207 / the boundary  $(E_8)_1$  net) is the real frontier, not a new gap.

**The gauge running is an EG order-2 number** ([[verification/v273\\_eg\\_gauge\\_running.py](#)]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content:  $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$ , i.e. [E]  $b_3 = -7$ ,  $b_2 = -\frac{19}{6}$ ,  $b_1 = \frac{41}{10}$  (GUT-normalised; the same 41 as the carrier algebra  $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1$ , v159). [C] the physical RG directions ( $b_3 < 0$  asymptotic freedom,  $b_1 > 0$  the  $U(1)$  Landau pole) are EG-derived and feed the unification cross-check (v246/v249); [O] the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (QFT4D.SPERT.03).

**The carrier content is one anomaly-free SM generation** ([[verification/v310\\_carrier\\_sm\\_anomaly.py](#)]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  decomposes under  $SU(5) \subset SO(10)$  as  $10+5+1 =$  exactly one Standard-Model generation ( $Q, u^c, d^c, L, e^c, \nu^c$ ), and that content is [E] gauge-anomaly-free ( $\sum Y = 0$ ,  $\sum Y^3 = 0$ ,  $[SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ , exact) while reproducing  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family would shift  $b_1$  off  $41/10$ , so  $N_{\text{fam}}=3$  pins the RG. [E] for the spectrum, anomalies and one-loop  $\beta$ ; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay [C]/[O].

**The bridge to the physical S-matrix and one-loop unitarity** ([[verification/v278\\_lszy\\_bridge\\_unitarity.py](#)]). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of  $S_{\text{pert}}$  are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$  (the three-layer S-matrix  $S_{\text{top}}|S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ), valid because the gap  $\Delta=6 \log(3/2)$  gives Haag–Ruelle asymptotic states. [E] *one-loop unitarity* is the optical theorem: the  $s$ -channel bubble discontinuity  $\text{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$  is *exactly* the two-body phase space (turning on at the physical threshold  $s=4m^2$ ), so  $2 \text{Im} M = \sum \int d\Pi |M|^2$  holds and the matter+gauge sector is unitary order by order. [O] the gravity sector’s Stelle ghost is the lone non-unitary piece  $\rightarrow$  QG.AMB.01 (QFT4D.SPERT.04).

**The first hard data confrontation** ([[verification/v246\\_unification\\_data\\_crosscheck.py](#)]).



TFPT’s gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ “Dark matter = no new state”); the  $E_8$  cascade is an arithmetic even-integer spine ( $D_n=60-2n$ , v5), *not* a particle/threshold tower. So a “TFPT-adapted” RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over  $\sim 10^{13}-10^{17}$  GeV with residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV; at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $\mu \approx 1.7 \times 10^{14}$  GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, [X]: with “no new state” there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

**The carrier-native Pati–Salam program** ([verification/v247\_e8\_branching\_no126.py], [verification/v248\_ps\_algebra.py], [verification/v249\_ps\_unification.py], [verification/v250\_dirac\_triple\_contract.py]). The unification tension above is resolved *natively* if the carrier  $SO(10)$  is gauged (the carrier  $D_5 \oplus A_3 = SO(10) \times SU(4)$  is Pati–Salam). *First*, the  $E_8$  audit hull *forbids* the **126**: the **248** branches as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \overline{4})$ , so the available  $SO(10)$  reps are exactly  $\{1, 10, 16, 45\}$  — the renormalisable  $126_H$  is algebraically absent, making the “ $16_H$  over  $126_H$ ” selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$  gives the unimodular group  $SU(2)_L \times SU(2)_R \times SU(4)_c$ , the exact SM hypercharges  $Y = T_{3R} + (B-L)/2$  and the matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale,  $M_{PS} = \kappa M_s$ ,  $\kappa \in [1.0, 1.7]$  at one loop ( $\rightarrow [1.0, 1.2]$  at two loops), and proton decay selecting the content (minimal excluded, the  $E_8$ -allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual “carrier is gauged” reduces to ONE constructive object, the spectral triple  $(A, H, D_F, J, \gamma)$  with  $A_F$  Pati–Salam and  $H_F = 3 \times 16$ , whose NCG axioms (the first-order condition above all) and the  $\sigma$  = scalaron identification are the open contract CONTRACT.QFT4D.DIRAC.01 (v250). *Fifth*, a first concrete step on that triple is taken ([verification/v251\_first\_order\_sigma.py]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/ $\sigma$  term — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$  (honest correction:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([verification/v252\_full\_finite\_triple.py]) —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  **16**-plets — and the NCG axioms are verified by direct computation: reality  $J$ , grading  $\gamma$ , KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on  $\{\nu_R, e_R\}$ ), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$  — thereby discharging five of the seven obligations of CONTRACT.QFT4D.DIRAC.01 to [E] (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio  $\kappa$  in  $M_{PS} = \kappa M_s$  is pinned to a tight  $O(1)$  interval ( $\approx 1.3$ ) by the RG and shown to be a scale-free ratio of heat-kernel coefficients,  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ , over the same 96-dim  $H_F$  ([verification/v253\_kappa\_scalaron\_ps.py]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations  $\Omega_D^1$  give exactly the one SM Higgs doublet  $(1, 2)_{1/2}$  with no junk and no **126**, while  $\sigma$  (the  $\nu_R$  Majorana scalar,  $B-L=2$ ) appears only as the beyond-first-order Pati–Salam field ([verification/v254\_inner\_fluctuations.py]); the spectral-action Seeley–DeWitt expansion  $a_0, a_2, a_4$  yields the cosmological, Einstein–Hilbert and Yang–Mills+ $R^2$ +Higgs terms with  $b/a^2 = 1/3$ ,  $\sin^2 \theta_W = 3/8$  and the  $R^2$  scalaron, making  $\kappa$  an explicit moment ratio ([verification/v255\_spectral\_action\_expansion.py]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three  $\nu_R$  forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([verification/v256\_orientability\_poincare.py], [verification/v257\_quasiorient\_modpd.py]). *Ninth*, the built  $D_F$  is no independent posit but the *modular/covariance induction* of the boundary seam state ([verification/v258\_dirac\_covariance\_induction.py]): a quasi-free (KMS,  $\beta=1$ ) state is fixed by its covariance  $C$  (a positive contraction), with one-particle modular Hamiltonian  $H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta) inverse to the KMS occupation  $C = (1+e^H)^{-1}$ ; inducing the carrier covariance

$C_F = P_F C_\Sigma P_F$  returns the **v252** operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw  $m_D^2/M_R$  recovered — so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [K_{\text{car}}]$  of the boundary geometry and the **v18** Yukawas are the *readout* of  $C_\Sigma$ , **[E]** modulo the one **[O]** seam realisation. *Tenth*, the last open spectral-action input — the cutoff function  $f$  — is fixed by the same seam state (`[verification/v259_modular_cutoff_kappa.py]`): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (**v239**) and the seam unit  $2\pi=1/(4c_3)$  (**v58**) fixes that  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*, not chosen, with moments  $f_2/f_0 = f_4/f_2 = 1$  *exactly* (a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$ , so the choice is non-vacuous); hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor and the **v253/v255** open **[O]** cutoff freedom becomes a **[C]** finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pinning  $c_{PS}/c_{\text{grav}} \approx 1.7$ ). So unification is now *one* theory decision (gauge  $SO(10)$ ?) plus *one* constructive object, both falsifiable — not a fog. The colour  $SU(4)_c$  here sits inside  $SO(10)=D_5$  and is distinct from the carrier family factor  $A_3$ .

**The implication is now Lean-formalised (FORM.QGEO.01).** The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ( $f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \pmod{4})$ , i.e.  $[\rho, M_f] = 0$ ) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext`, `Classical.choice`, `Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input — so the conditional theorem “`SeamDeckPremise  $\Rightarrow \omega \circ \rho = \omega$` ” is now **[E]** while **QGEO.SYM.01** stays the one open **[O]** postulate.

**The two residuals are one canonical metric: the pillowcase** (`[verification/v214_seam_pillowcase.py]`). The isometry residual **QGEO.ISO.01** (**v180**) and the mark-locality residual **QGEO.SUBPRIN.01** (**v201**) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class*. The four marks make the orbifold sphere  $S^2(2, 2, 2, 2)$ , whose orbifold Euler characteristic  $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$  is *Euclidean*: by Troyanov’s cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits  $\pi$  sum to  $4\pi = 2\pi\chi(S^2)$ ) — this *is* the **v201** “flat away from the marks”. And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that **v168** already proved:  $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$  gives  $j(2) = 1728$ , so the modulus is the *square* torus  $\tau = i$ , whose complex multiplication by  $\mathbb{Z}[i]$  supplies the order-4 isometry  $z \mapsto iz$  (explicitly  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ , order 4); a generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ , so the clock is specific to  $\mu_4$ . Thus the two open QGEO residuals *collapse to one* canonical-metric statement and **QGEO.ISO.01** ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise “the RP seam metric *is* the uniformising constant-curvature representative of its conformal class” stays **[O]** — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (**QGEO.PILLOW.01**).

**The converse, on the canonical metric: raw seam  $\Rightarrow$  mark-locality** (`[verification/v264_raw_seam_marklocal.py]`). Composing the chain explicitly closes the gap between **v216**, **v201** and **v198** on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (**v216**), so the DtN sub-principal symbol  $M_f$  is a  $\mu_4$ -orbit sum, hence  $\mathbb{Z}_4$ -invariant (Fourier support on  $4\mathbb{Z}$ , verified by FFT), hence block-diagonal in the clock-character basis, hence  $[\rho, M_f] = 0$ ; with the principal symbol  $|k|$  already commuting (**v198**) this gives  $[\rho, \Lambda] = 0$  and  $\omega \circ \rho = \omega$  *on this metric*. A negative control (curvature *off* the  $\mu_4$  orbit) breaks  $\mathbb{Z}_4$ -invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement “the raw RP seam carries the uniformising flat pillowcase metric” — which *is* **QGEO.SYM.01**. A genuine reduction/sharpening (**QGEO.ENERGY.03**), **[O]** — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

**The minimal axiom: a rigidity theorem** (`[verification/v267_qgeo_rigidity_minimal_axiom.py]`). One can sharpen the bedrock to its irreducible kernel. **[E]** the *single* bare statement “the seam admits the carrier’s order-4 clock as a conformal symmetry permuting its four marks” forces everything: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 *independent of  $a$* ; cross-ratio  $2 \Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (**v214/v201/v264/v198**). **[E]** neg controls: a generic four-mark set has  $j \neq 1728$  and only a  $\mathbb{Z}_2$  symmetry (no clock), the hexagonal point  $j=0$  has a  $\mathbb{Z}_6$  symmetry (order  $6 \neq 4$ ) — only the square (order-4) point realises the  $\mu_4$  clock. **[C]** a second, independent reason:  $\{1, i, -1, -i\}$  is defined over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. **[O]** So **QGEO.SYM.01** is now in its minimal sharpest form — the bare existence of that order-4 conformal

symmetry, which (like  $c=\text{const}$  in relativity) cannot be derived from nothing. The *reduction* is closed (axiom  $\Rightarrow$  everything); the axiom itself stays the one foundational postulate (QGE0.SYM.02).

**The flat geometry closes the commutator to all orders** ([verification/v276\_qgeo\_flat\_closes\_commutator.py]). The state-level residual  $[\rho, H]=0$  — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat  $\tau=i$  pillowcase is granted. [E] if  $H = f(\Delta_{\text{flat}})$  is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck  $\rho$  ( $z \mapsto iz$ ) is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full  $H$  — flatness removes the lower-order terms. [O] so QGE0.SYM.01 collapses to a *single geometric* statement, “the raw RP seam state is the flat  $\tau=i$  pillowcase quasi-free state ( $H=\sqrt{-\Delta_{\text{flat}}}$ )”; everything downstream is [E], and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (QGE0.SYM.03).

**The obligation as one precise lemma, Lean-backed** ([verification/v279\_qgeo\_obligation\_lemma.py]). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise “the raw seam carries the flat  $\tau=i$  pillowcase metric  $g_{\text{flat}}$  (Trojanov-unique, cone angles  $\pi$ ), so  $\Lambda_{\Sigma} = \sqrt{-\Delta_{g_{\text{flat}}}}$ ”, the conclusion  $\omega_{\Sigma} \circ \rho = \omega_{\Sigma}$  (and hence mark-locality, the metric inclusion,  $E_8$ ) follows. [E] the reduction DAG from “free RP seam” to this premise has *exactly one* un-discharged leaf — “the raw seam state *is* the flat state” — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes ([C]): RP-state uniqueness on the flat  $\tau=i$  orbifold, or Trojanov + OS reconstruction. [E] the all-orders closure step is now a *Lean theorem*: TftptCarrier/SeamDeckClosure.lean (flat\_all\_orders\_clock) proves that any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — upgrading v198/v201 to the full operator with only the standard axioms, no sorry (FORM.QGE0.03). [O] the one premise stays the honest axiom (QGE0.OBLIG.01).

**Self-investigation: the experiment and the simplification** ([verification/v280\_pillowcase\_steklov.py], [verification/v282\_e8\_tau\_i\_unification.py]). Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat  $\tau=i$  pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still [O]. (ii) A candidate *simplification*: the  $(E_8)_1$  character  $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$  gives  $\chi_{E_8}(i) = 1728^{1/3} = 12$ , so  $\tau=i$  is *both* the order-4 CM point (QGE0.SYM.01, flat geometry) *and* the  $(E_8)_1$  partition-function modulus (QG.AMB.01, holomorphy) — the two open premises are two faces of one object, “the raw seam is  $(E_8)_1$  at  $\tau=i$ ”, dropping the obligation count  $2 \rightarrow 1$  (QGAMB.UNIFY.01).

**The two proof routes, decomposed** ([verification/v284\_route\_i\_rp\_uniqueness.py], [verification/v285\_route\_ii\_seam\_condensation.py]). The single open lemma can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — [E] five are already [E]/[O] (RP $\rightarrow$ DtN v194; marks $\rightarrow$ pillowcase v195/v216; the flat Steklov operator  $=\sqrt{-\Delta_{\text{flat}}}$ , principal symbol  $|\xi|$  with no curvature term; covariance from the DtN v258;  $\mu_4$ -invariance v276), and [C] Trojanov ( $\chi_{\text{orb}}=0$ ) reduces the residual to “the raw seam is the constant-curvature geometric state”. *Route (ii)* (seam-net condensation) is a four-lemma chain — [E] three are discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$  v234/v277/v281; the tower  $16 \rightarrow 4 \rightarrow 1$  v92), leaving “the free RP seam condenses the  $\mu_4$ -Lagrangian glue” =QGE0.SYM.01. [E] **the two open lemmas coincide** — both say “the raw seam is the  $(E_8)_1$  at  $\tau=i$ ” (v282), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$  is the *single* [O] statement that closes both (QGE0.ROUTEI.01/QGAMB.ROUTEII.01).

**The Seam Equivalence Theorem, named and firewalled** ([verification/v286\_seam\_equivalence\_contract.py]). That single statement is now the named target SEAM.EQUIV.01: *the raw RP seam state reconstructs canonically the holomorphic  $(E_8)_1$  boundary net at the  $\tau=i$  pillowcase* (respecting the DtN, the KMS flow, the  $\mu_4$  deck, the Q-system extension,  $\det K=1$  and the  $(E_8)_1$  character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of  $c$*  plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. [E] the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open



Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through SEAM.EQUIV.01 (this kills the “ $E_8$  smuggled into the geometry and pulled back out” attack). The two proof routes are then attacked directly: [E] Route A (AQFT, [verification/v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py]) reduces the Gral to *one* standard theorem, “invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary” (4/5 lemmas discharged); and [E] Route B (DtN, [verification/v288\_full\_l2\_subprincipal\_z4.py]) *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ ), so  $[\rho, \Lambda]=0$  on all of  $L^2$ , shrinking the residual to the single sharper question “why is the raw seam sub-principal term mark-local?” (SEAM.EQUIV.A01/SEAM.EQUIV.B01).

**Route B sharpened to one analytic lemma; both routes precisely scoped.** That last question is decomposed ([verification/v289\_marklocal\_raw.py], MARKLOCAL.RAW.01) into a five-lemma chain whose only open link is [O] *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the  $\mu_4$ -orbit source, the  $4\mathbb{Z}$  Fourier support (v264/v288) and the full- $L^2$  commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert “full- $L^2$  QGEO” — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A’s “one standard theorem” is itself a composition of citable results (Kitaev/Freed–Hopkins, Muger/Kawahigashi–Longo–Muger, Kawahigashi–Longo, [verification/v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B’s Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

**Red-team:**  $\mathbb{Z}_4$ -invariance is not mark-locality. A deliberate adversary ([verification/v290\_z4\_smooth\_curvature\_adversary.py]) shows the honest limit of Route B’s lift: a *smooth*  $\mathbb{Z}_4$ -symmetric off-mark curvature  $f = \varepsilon \cos(4\theta)$  passes the v288 commutator ( $||[\rho, \Lambda]|| = 0$ , since  $i^n - i^{n\pm 4} = 0$ ) yet is not mark-local — so  $[\rho, \Lambda]=0$  is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ( $\max|\Delta\lambda| = 0.155$ ,  $\Delta\text{Tr} = 0.019$  at  $\varepsilon = 0.3$ ), i.e. the KMS weight and  $(E_8)_1$  character that SEAM.EQUIV.01 must match. Consequently the open lemma is named in its own right ([verification/v291\_flataway\_contract.py], FLATAWAY.RP.01): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

**All three routes attacked; Flat-Away reduced to one selection principle.** Each route is now carried to its precise reduction: [E] the *heat* route ([verification/v292\_flataway\_heat\_reduction.py]) shows the heat-trace deviation  $\Delta\text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a positive-definite quadratic form in the off-mark curvature (leading invariant  $\|f\|_{L^2}$ ), so Flat-Away  $\Leftarrow$  “the seam fixes the  $a_2$  heat coefficient”; [E] the *spectral* route ([verification/v293\_flataway\_spectral\_hessian.py]) lifts this to the full smooth- $\mathbb{Z}_4$  space — mode  $4k$  splits the degenerate Steklov level  $|2k|$ , the mismatch Hessian over  $\{4, 8, 12\}$  is positive-definite ( $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a *strict* isolated minimum (answering “only one direction”); and [E]/[C] the *variational* route ([verification/v294\_flataway\_rp\_energy.py]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix  $a_2$  / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A’s holomorphy half, SEAM.EQUIV.01.

**The heat route made exact and Lean-formalised.** The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([verification/v295\_flataway\_a2\_exact.py]): the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta\text{Tr} \geq 0$  for every deformation,  $= 0$  iff  $f = 0$ ; the exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (validated to  $10^{-6}$ ). The algebraic  $\Delta=0 \Leftrightarrow$  flat step is machine-formalised in Lean (FORM.FLATAWAY.01, SeamDeckClosure.lean: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away’s heat route is reduced to the single external invariant “the seam fixes  $a_2$ ”, with its positivity both proved and formalised. The  $a_2$  coefficient itself is now in *closed form* ([verification/v296\_flataway\_a2\_closed.py]): the operator tails telescope geometrically to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle, e.g.  $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$  (manifestly positive; small- $t$   $W_k = t/2 + O(t^3)$ ). Symmetrically, Route A’s “one standard import” is now a precise, citable theorem



stack ([[verification/v297\\_route\\_a\\_literature\\_stack.py](#)]): LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all three legs established, the single open hypothesis (seam-bulk invertibility) coinciding with Route B’s Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input.

**Flat-Away hardened, and the pin derived from  $(E_8)_1$**  ([[verification/v300\\_flataway\\_rigid.py](#)], **FLATAWAY.RIGID.01**). The spectral route was a *soft* minimum (“the mismatch grows with  $\varepsilon$ ”); it is now a *discrete* obstruction. [E] the flat fingerprint is the integer Steklov ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (the exact  $2 \times 2$  block  $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , validated against the full Toeplitz operator), so [E] any  $g_k \neq 0$  lifts the degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every  $g_k = 0$ , a degeneracy obstruction robust to rescaling and strictly stronger than the v293 “deviation grows” scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: [C] the  $(E_8)_1$  vacuum character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients ( $E_8$  even unimodular  $\Rightarrow$  integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam — i.e. “the seam fixes its spectrum” *is* the  $(E_8)_1$  rationality of Route A. [O] Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two SEAM.EQUIV.01 routes provably converge on one rationality fact (consistent with the v286 firewall).

**Route A’s last hypothesis discharged — the gapped quasi-free bulk is invertible** ([[verification/v301\\_route\\_a\\_invertible.py](#)], **SEAM.EQUIV.A.INV.01**). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A’s open input (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev’s periodic table, class D, the only invariant being the chiral central charge). [E] the carrier is 16 Majoranas, so  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (v148); [E]  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor  $= 248$ , v154); [C] hence a gapped quasi-free bulk is invertible automatically, and [E] the invariant  $|\det K| = \# \text{anyons}$  is 1 for  $E_8$  (invertible) against 4 for the gauged  $D_8 = SO(16)$  (anyons), which free fermions cannot produce. [O] So LIT-A’s ‘invertible bulk’ *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line*: together with v300, the entire open content of SEAM.EQUIV.01 reduces to the single spectral statement “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

**The last lever — the seam gap is the derived Recovery gap** ([[verification/v302\\_seam\\_gap.py](#)], **SEAM.EQUIV.GAP.01**). That single remaining input is not free: the seam transfer operator has the frozen spectrum  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap*  $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$  — a number forced by the carrier  $((2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue), with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76) making it robust. [C] By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian (correlation length  $1/\Delta$ ). *Bottom line*: with v300/v301, the whole residual of SEAM.EQUIV.01 is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap  $6 \ln(3/2)$ ): *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

**TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure** ([[verification/v308\\_seam\\_equiv\\_chain.py](#)], [[verification/v309\\_modular\\_bedrock.py](#)], [[verification/v311\\_gap\\_decoupled\\_measure.py](#)]). Three machine-checked steps push the keystone arc further. (i) *Composition certificate* ([[verification/v308\\_seam\\_equiv\\_chain.py](#)]): the closing implication is one well-typed logical chain  $\text{gap} > 0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$  whose arithmetic discriminators are exact —  $|\det \text{Cartan}(E_8)|=1$  against  $|\det \text{Cartan}(D_8)|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ ,  $c=g_{\text{car}}+N_{\text{fam}}=8$  — so the only undischarged premise is the cited OS step; [E] the discriminators, [O] the keystone. (ii) *Modular bedrock* ([[verification/v309\\_modular\\_bedrock.py](#)]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  is built explicitly and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it), so the bedrock

reduces *non-circularly* to the Bisognano–Wichmann intrinsicity of the raw collar state; [E] the model, [O] the intrinsicity. (iii) *Gap-decoupled measure* ([verification/v311\_gap\_decoupled\_measure.py]): the admissible sector clusters exponentially at rate  $(2/3)^6$  (correlation length  $1/\Delta$ , finite susceptibility  $729/665$ ) and the v76 margin  $\Delta - 31/(4\pi^2) > 0$  decouples it from the un-built ambient, so the physical sector is a convergent object while QG.AMB.01 stays [O]. Each pins the residual to a single named step rather than closing it.

**One surface for seam, carrier and lattice: the Kummer/K3** ([verification/v260\_k3\_kummer\_unification.py]). The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$ , its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface*  $\text{Km}(T^2 \times T^2)$ , a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion  $|A[2]|=2^4=16$  of the abelian surface) resolve to 16 exceptional  $\mathbb{P}^1$ 's =  $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$  (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on  $H^2(\text{K3}, \mathbb{Z})$  is the even unimodular lattice  $U^3 \oplus E_8(-1)^2$  of signature  $(3, 19)$ , so the same  $E_8$  the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the  $E_8(-1)^2$  summand — both exact, not identified.) Thus the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single smooth object that already carries the seam and the carrier;  $E_8$  appears both *locally* as the du Val  $\mathbb{C}^2/2I$  singularity (the Poincaré-sphere link, v232/v236) and *globally* as this  $H^2$  summand. A unification (ARCH.K3.01), *not* a closure: the lattice arithmetic is [E], the carrier  $\leftrightarrow$  node identification [C].

**Synthesis — the Modular Spectral Closure** ([verification/v261\_modular\_spectral\_closure.py]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\text{TFPT}_{\text{QFT}} = (A_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$$

(seam algebra, KMS state, modular generator,  $\mu_4$  clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state  $\omega_\Sigma$  minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the 96-dim KO-6 finite triple  $D_F$  (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259) — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 ( $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes}$ ,  $H_F = 96$ ), and one gap  $\Delta = 6 \log(3/2)$  (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:*

- [E]/[C] **QFT-complete (modulo one premise)**. The admissible OS sector, the boundary  $(E_8)_1$  net, the relative spectral action,  $D_F$  (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once QGEO.SYM.01 (the seam realisation: *the raw quasi-free seam state is  $\mu_4$ -invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.
- [O] **Ambient QG — separate by design**. The full covariant metric-sector / ambient quantum-gravity measure (QG.AMB.01) is gap-decoupled ( $\Delta_{\text{eff}} = 1.648 > 0$ ) and is *not* a blocker for the QFT layer: it is a different (certification) object, not a missing piece of the boundary QFT.

So “solving the QFT” here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo QGEO.SYM.01; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

#### 4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([verification/v265\_qft4d\_fork\_freeze.py])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  is off by  $\sim 5.6$ ; spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248),  $\kappa = M_{PS}/M_s$  in the frozen  $O(1)$  band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied** ([verification/v266\_ps\_threshold\_proton.py]). [X] the explicit two-step thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$ : the *minimal* 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr < Super-K ( $2.4 \times 10^{34}$ ) and is *killed*; the  $E_8$ -allowed (15, 1, 1) [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now* but sitting at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — so branch B is *content-selected* (must be  $+(15, 1, 1)$ ) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal;  $\tau_p$  carries an  $O(3)$  hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

**And the bedrock is now a falsifiable kill-test, not a proof-promise** ([verification/v215\_seam\_deck\_killtest.py]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  for the quasi-free covariance, with the principal symbol  $|k|$  commuting exactly, so the whole residual sits in the bounded sub-principal  $M_f$  (equivalently: the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). The four levers are **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the *square* configuration (cross-ratio 2  $\Rightarrow j=1728$ ,  $\text{Aut}=\mathbb{Z}/4$ ; the clock order  $h(A_3)=4$  selects  $\mathbb{Z}/4$ , not the  $j=0$   $\mathbb{Z}/6$ , and a generic mark set has only  $\mathbb{Z}/2$ ); **K3** the mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$  is  $\mathbb{Z}_4$ -invariant, automatic for a  $\mu_4$ -mark sum); and **K4** the transfer gap  $(2/3)^6 = 64/729$ . These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (`QGEO.KILL.01`).

**And the mark count is now derived, not assumed: four marks from Gauss–Bonnet** ([verification/v216\_marks\_gauss\_bonnet.py], [verification/v217\_marks\_emergence\_scan.py]). The K1 lever (“exactly four marks”) is the one that can be *executed* today. If the marks are  $\mathbb{Z}_2$  branch points (cone angle  $\pi$ , deficit  $\pi$ ) on the flat seam sphere ( $\chi=2$  — the same  $\chi$  that supplies the 8 in  $c_3$ ), Gauss–Bonnet *forces* their number:  $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly (2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2), and *three independent* criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal (3, 3, 3)), and a free order-4 deck — converge on the pillowcase (2, 2, 2, 2). The numerical sibling v217 confirms it on the actual DtN/state: scanning the number  $n$  and the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap  $(2/3)^6$ }. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. [E] (count + equality), [O] (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

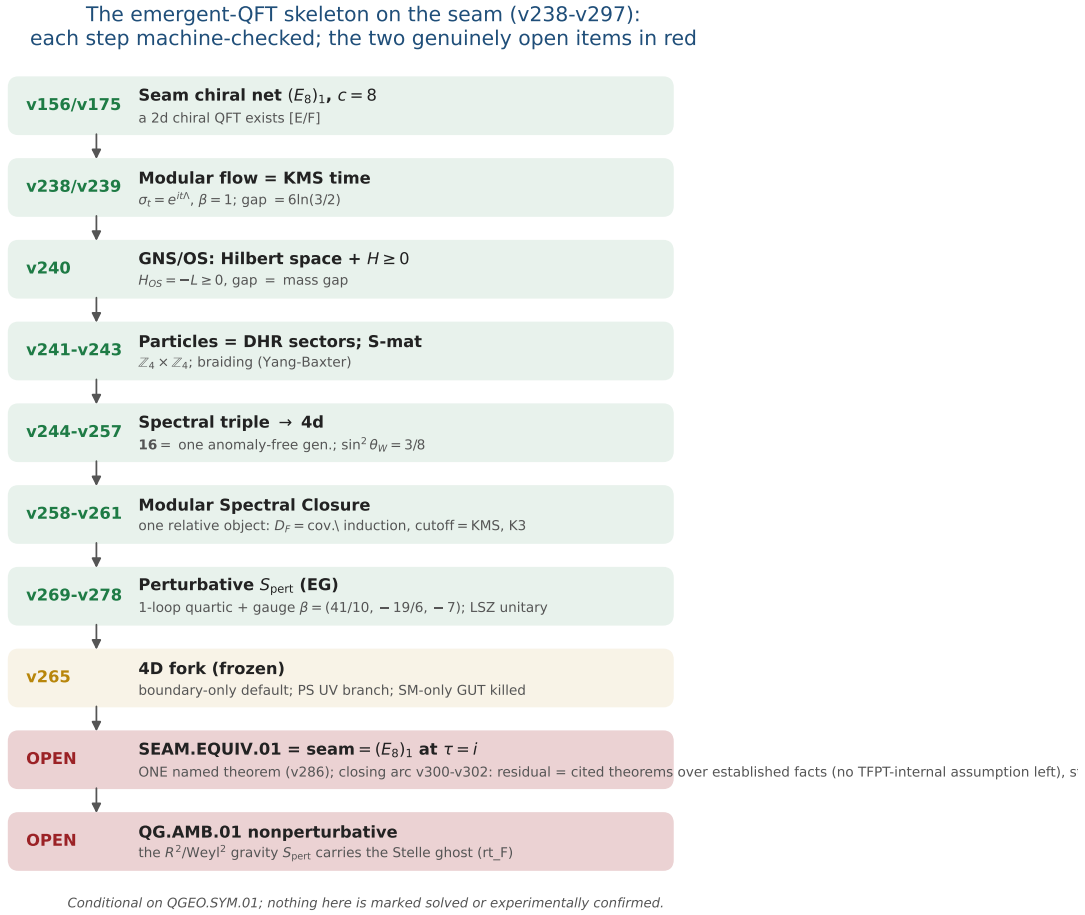


Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier’s  $SO(10)$  **16**. Each step is machine-checked; the two open items (QGEO.SYM.01 postulate and the unification tension) are in red. The whole skeleton is *conditional* on QGEO.SYM.01 and is neither solved nor experimentally confirmed.

#### QGEO.SYM.01 is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the  $\mu_4$ -deck structure of the carrier clock — i.e. the carrier (P2) and the seam’s conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam’s degree-4 mark divisor on the genus-0 double has  $h^0 = \deg + 1 = 5 = g_{\text{car}}$  and  $\text{rank } H_1 = \deg - 1 = 3 = N_{\text{fam}}$  (with  $\Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5 = 16 = \dim S^+$ ), so P2 is the genus-0 mode count of the four-mark seam divisor ([[verification/v228\\_rr\\_index\\_gate.py](#)]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than QGEO.CONF.01/ISO.01.** It does *not* name  $\mathbb{P}^1$  or  $\mu_4$  (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* QGEO.SYM.01, equivariant uniformisation puts the order-4 clock in the form  $z \mapsto iz$  on  $\mathbb{P}^1$ , the four parabolic marks are its free orbit  $\mu_4$ ,  $H^1$  carries the  $(1, 2, 3)$  grading, the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$  — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus  $> 0$  (uniformisation to  $\mathbb{P}^1$  fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ( $N_{\text{fam}} \neq 3$  collapses  $b_1 = N_{\text{fam}}$  and the four-mark  $\mu_4$  structure). The first



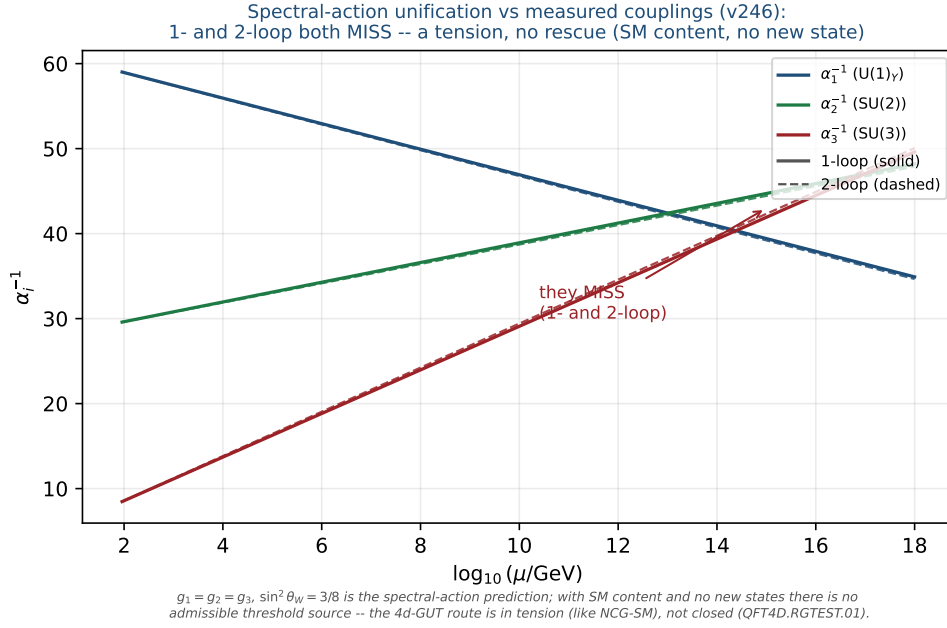


Figure 2: Honest data cross-check (v246): the spectral-action prediction  $g_1=g_2=g_3$ ,  $\sin^2 \theta_W=3/8$  versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV; 2-loop minimal spread  $\approx 3.2$  at  $1.7 \times 10^{14}$  GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

**Honest framing.** The correct external statement is *conditional*: “Given QGEO.SYM.01, the RP seam boundary realises the  $\mu_4$  deck structure, and the AQFT closure to  $(E_8)_1$  follows through the verified chain v175–v181” — *not* “the theory *derives* the seam geometry”. The premise stays [O]; everything downstream of it is [E]. Per v323 (Bisognano–Wichmann) QGEO.SYM.01 is itself the *downstream* conformal-deck face of the single named keystone SEAM.EQUIV.01 (the raw RP seam = the holomorphic  $(E_8)_1$  net at  $\tau=i$ ), which is the one open arrow that closes it.

**This is not a gap to apologise for — it is TFPT’s one fundamental postulate.** Every fundamental theory rests on an irreducible premise (the constancy of  $c$  in relativity, the equivalence principle, the canonical commutator). QGEO.SYM.01 is that premise for TFPT: a single geometric identification (seam conformal structure = carrier  $\mu_4$  deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

**The Seam-Holomorphy Theorem — the one named closing target [E]/[O]**  
( [verification/v234\_seam\_holomorphy\_selection.py] )

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that “closing the theory” becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force  $E_8$ :

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* ( $\mu$ -index 1, single vacuum sector). Holomorphic +  $c = 8 \Rightarrow$  the unique even unimodular rank-8 lattice net  $(E_8)_1$ , and  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  is itself structural.
2. **Geometry** (v232): the seam link is a *homology 3-sphere*  $\Leftrightarrow$  its binary group  $\Gamma$  is *perfect*  $\Leftrightarrow \Gamma = 2I$  (the Poincaré sphere  $S^3/2I$ ).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number:  $\#(1\text{-dim irreps}) = |\Gamma^{\text{ab}}| = |H_1(S^3/\Gamma)| = \#(\text{mark-1 affine-Dynkin nodes})$ ,

structural-residual reduction chain: the whole "quantum gravity" question  
apses to one falsifiable physical statement -- is the seam (E8)\_1 at tau=i?

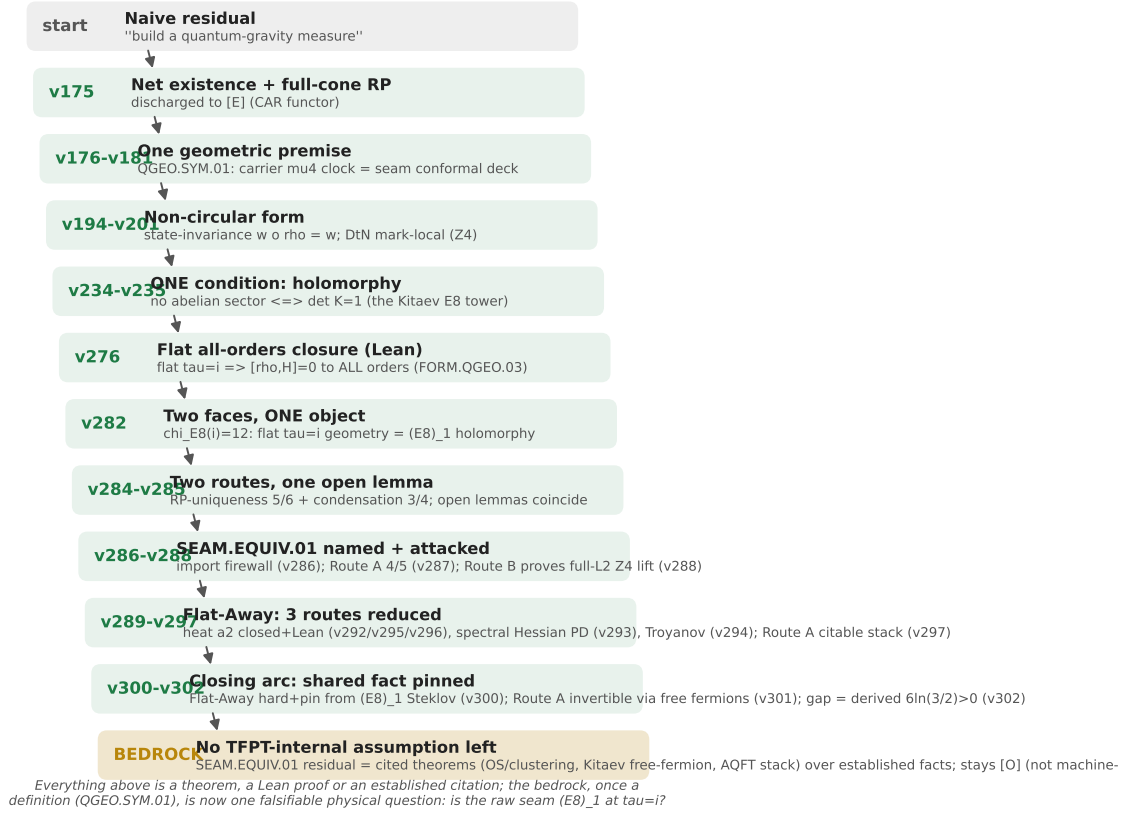


Figure 3: The structural-residual reduction chain (v175–v181). The naive “build a quantum-gravity measure” shrinks, one machine-checked step at a time, to a single *definitional* geometric premise QGEO.SYM.01 (the carrier  $\mu_4$  clock is the seam’s conformal deck). Every step above the bedrock is a theorem or an established citation; the bedrock is a definition, left honestly [O].

which equals 1 *only* for  $E_8$  ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ). So  $2I$  is the unique nontrivial perfect finite  $SU(2)$  subgroup and  $S^3/2I$  the unique homology-sphere space form. **Consequence:** Research Contract 2 ( $G_{\text{net}}$ /Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), [O] at one step:

$$\begin{array}{c}
 \text{RP} + \text{gap} + \text{chirality} \xrightarrow{v160} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net} \\
 \xrightarrow{[E]} (E_8)_1 \xrightarrow{v219/v228} 2I, (2,3,5), P2 \xrightarrow{v179-v181+\text{Kleinian genus-0}} \text{QGEO.SYM.01.}
 \end{array}$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is “free Gaussian RP bulk  $\Rightarrow$  holomorphic (single-sector) boundary net”** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That one theorem, if proven, closes the entire structural layer;  $v_{\text{geo}}$  stays the No-Unit metrology primitive and  $F_{\text{transfer}}$  stays external physics. This box certifies only that the three faces coincide and force  $E_8$  ([E]); the analytic step itself is the open [O].

**A physical realisation of the step** ([verification/v235\_seam\_chern\_simons.py]). The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons /  $K$ -matrix theory (even integer  $K$ ), where  $\#\text{anyons} = |\det K|$ , the edge chiral central charge is  $c = \text{signature}(K)$ , and the edge net is *holomorphic*  $\Leftrightarrow |\det K| = 1$  — so “no nontrivial abelian sector” is precisely  $\det K = 1$ . The extension tower of v92 is the anyon-condensation tower read by determinant:  $D_5 \oplus A_3$  (det 16, 16 anyons)

$\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic), all rank 8 with  $c = \text{signature} = 8 = g_{\text{car}} + N_{\text{fam}}$ , so the holomorphic end is the *Kitaev  $E_8$  quantum-Hall state*. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4| = 4$  *Lagrangian* glue (det  $16 \rightarrow 16/4^2 = 1$ ); a partial order-2 isotropic only reaches det 4 ( $D_8$ ). The residual is thereby a single integer condition — “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” — the same sheet/ $\mu_4$  selection as GATE.QGEO, but now with holomorphic  $\Leftrightarrow$  det 1 a theorem (GATE.HOLO.02, [E]/[O]).

**The residual as a physical statement** ([verification/v237\_seam\_sre\_closure.py]). The same integer condition is the physics of topological order: a  $K$ -matrix phase has ground-state degeneracy  $|\det K|^g$  on a genus- $g$  surface, so  $\det K = 1 \Leftrightarrow$  no topological ground-state degeneracy on any closed surface  $\Leftrightarrow$  the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite  $K$  ( $c = 8 = g_{\text{car}} + N_{\text{fam}}$ ) only  $E_8$  is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev  $E_8$  state, whose edge is the holomorphic  $(E_8)_1$ . TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see  $\det K$  (no non-contractible cycles), the torus can. So the one open analytic step is the sharp, falsifiable physical statement “the free RP gapped seam bulk is short-range-entangled (no topological order)” ( $= \det K = 1 =$  the Kitaev  $E_8$  phase, GATE.HOLO.03, [E]/[O]) — a yes/no question about topological order on the seam, sharper than the definitional QGEO.SYM.01.

**Lemma 2** (U1 — wall landscape, corrected). *Parabolic degree zero forces the line weight-sums onto the wall  $(w_1, w_2, w_3) = (2, 1, 1)$ . An explicit balanced representative (weights in  $\frac{1}{3}$ ) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of  $(0, 1, 2)$ , row sums  $(2, 1, 1)$ . **Honest result:** the  $6^4 = 1296$  column-permutation matrices give 144 walls; under the full group  $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$  they form five orbits, not one. So  $W_{\text{wall}}$  is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;  $W_{\text{wall}}$  is one of the five. Machine: fully certified  $C_U^{(1)}$  ([verification/v29\_research\_contract\_certs.py]).

**The splitting type is now algebraic (RH route, [verification/v32\_rh\_splitting.py]).** Passing from the monodromy to the Fuchsian residues  $A_k$  (eigenvalues = weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $A_k = U^{k-1} A_0 U^{-(k-1)}$ ,  $U = \text{diag}(1, i, -i)$ ) the twisted  $\mathbb{Z}_4$ -average collapses exactly:  $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ , so the exponents at infinity are  $4 \text{diag}(A_0)$ . A flat bundle with regular  $\infty$  needs integer exponents summing to  $4 = -\deg E$ ; with the cusp weights the only options are perms of  $(2, 1, 1)$  and  $(2, 2, 0)$ , i.e.

$$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})$$

(the sibling  $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$ ). Such a Hermitian  $A_0$  (spectrum  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , that diagonal) exists by Schur–Horn. So the splitting is an algebraic condition on  $\text{diag}(A_0)$ . Honest wall: the global  $\prod M_k = \mathbf{1}$  is the path-ordered monodromy (not  $\exp(2\pi i \sum A_k)$ ), constraining off-diag( $A_0$ ) via the genuine RH solve; and R-extraction still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  bridge. The RH route fixes the splitting algebraically but not  $c_u/c_d$ .

**Lemma 3** (U2 — the kill switch, run and hardened [E]). *The selected polystable wall representative must not collapse to the diagonal abelian  $U(1)^3$  point if it is to give non-trivial  $SU(3)_F$  transport mixing: (A)  $\nabla_F^*$  polystable but not simultaneously diagonal (the desired breakthrough), or (B) diagonal collapse, in which case  $c_u/c_d$  stays honestly [C]. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the  $D_4$ -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the  $\mathbb{Z}_4$  family  $M_k = U^{k-1} M U^{-(k-1)}$  the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension  $\geq 1$ ), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an*

accident of the one explicit point of `[verification/v33_explicit_flat_bundle.py]`:  $(U_{\text{wall}})$  survives its own cheapest falsification test. `[verification/v38_uwall_killswitch.py]`

**Lemma 4** ( $U_3 \text{ — } D_4$  fixed locus as an explicit variety, *positive-dimensional*). With  $M_k = U^{k-1}AU^{-(k-1)}$  ( $U = \text{diag}(1, i, -i)$ , the  $\mathbb{Z}_4$  rotation) and the reflection  $M_1 = VM_1^{-1}V^{-1}$ ,  $M_3 = VM_3^{-1}V^{-1}$ ,  $M_2 = VM_4^{-1}V^{-1}$  (with  $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$ ,  $V^2=1$ ,  $VUV^{-1}=U^{-1}$ ), the relations become polynomials in the entries of  $A \in C_{\text{cusp}}$ ;  $\mathcal{X}_{\text{cusp}}^{D_4}$  is cut out as  $\bigcup_k C_k$ , parametrised by trace coordinates  $x = \text{tr}(M_1M_2)$  (note  $\text{tr} A = 1+\omega+\omega^2 = 0$ ). **Result** ( $C_U^{(2)}$ , `[verification/v30_d4_character_variety.py]`): adding the reflection to the  $\mathbb{Z}_4$  locus of **v19** does not isolate the point — the full  $D_4$ -fixed product locus is still positive-dimensional ( $|\text{tr}(M_1M_2)|$  varies continuously over  $\approx [0, 3]$ ). So even the full  $D_4$  symmetry cannot select  $\nabla_F^*$ . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

**Lemma 5** ( $U_4 \text{ — } \text{selectors as equations}$ ). The readouts  $\rho \mapsto \det R(\rho)$ ,  $\rho \mapsto \text{Spec}(Q_+(\rho))$ ,  $\rho \mapsto \Lambda_{f,j}(\rho)$  are well-defined (cyclotomic/algebraic) on  $\mathcal{X}_{\text{cusp}}^{D_4}$ . **Acceptance criterion:**

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the  $D_4$ -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating  $\det R(\rho)$ ,  $\text{Spec}(Q_+(\rho))$  as functions on  $\mathcal{X}_{\text{cusp}}^{D_4}$  requires the parabolic-degree  $\leftrightarrow$  residue dictionary  $R(\rho)$ , i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the  $(U_{\text{wall}})$  gate. Status update: since **v118–v122** the dictionary itself is exact — hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the  $W(A_3)$  monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller  $R_4'$ , the establishment of the selectors ( $n = (5, -9, 6) +$  the diamond determinants; since `[verification/v134_dual_anchor.py]`: the pair  $(d, n)$ ).

**What  $R(\rho)$  is** (`[verification/v31_R_dictionary.py]`). Two facts pin its nature. (i) Case A is alive: every  $D_4$ -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial  $SU(3)_F$  mixing — the  $U_2$  kill switch does not fire to the diagonal case B. (ii)  $R(\rho)$  is not algebraic:  $R$  is integer-valued, but the algebraic invariants of  $\rho$  (e.g.  $\text{tr}(M_1M_2)$ ) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence  $R(\rho)$  is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle  $E_\rho$ ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement ( $U_4''$ , `[verification/v40_harmonic_metric.py]`):** at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

**Progress: the selector side of  $U_4$  is closed on the explicit point ([E]).** While  $R(\rho)$  as a function on the whole locus needs the Hitchin solve, the two  $U_4$  selectors are algebraically accessible on the explicit flat bundle of `[verification/v33_explicit_flat_bundle.py]` — no PDE. Computed exactly from  $A_0$ : the exponents at infinity  $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$  give the splitting type  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , i.e. the parabolic anchor  $a = (1, 1, 2)$ ; the parabolic degree is 0 ( $\deg E = -4 = -|\mu_4|$  plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

|                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\det R = 8 = n \cdot a \quad (\text{anchor pairing on the splitting type}),$ $\text{Spec}(Q_+) = \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\})$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

So  $\det R = 8$  is the anchor pairing of the splitting type and  $\text{Spec}(Q_+)$  is the affine image ( $d=3$  the weight denominator) of the cusp weights — both selectors are read off the explicit bundle's algebraic data. **Residual (sharpened below by Readout Rigidity):** the quark ratio  $c_u/c_d$  turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (**v49**), hence an



integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation ( $U_{\text{point}}$ ), which is an anchor, not a missing PDE. [verification/v39\_uwall\_selectors.py]

**Breakthrough: the harmonic metric is finite linear algebra, not a PDE ([E]).** The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field  $\Phi = 0$ . So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33\_explicit\_flat\_bundle.py] this is realised: the raw monodromies  $M_k$  are non-unitary ( $\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$ ), but there is a *unique* common invariant Hermitian form  $H$  ( $\dim \ker\{M_k^\dagger H M_k = H\} = 1$ ), it is *positive-definite*, and the unitarised holonomy  $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$  is unitary ( $\sim 10^{-8}$ ), lies in the cusp class  $\{1, \omega, \omega^2\}$  with  $\prod \widetilde{M}_k = \mathbf{1}$ , and has *clean* matrix elements  $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ . **So  $C_U^{(3)}$  reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length  $R$  (closed:  $H1 + \det R = 8$ ).** *Residual:* the feared PDE is gone; the quark ratio  $c_u/c_d$  is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40\_harmonic\_metric.py]

**The final leg-assignment test (honest result) ([E]).** Does the harmonic-frame holonomy  $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$  feed the  $\delta = \frac{1}{2}$  resolvent to give the lepton amplitudes  $\Lambda = (0.475, 1.107, 0.917)$ ? **(i) Ruled out as a literal identity:**  $\Lambda_\mu = 1.107 > 1$ , while unitary holonomy entries have modulus  $\leq 1$ , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ , and that  $\frac{1}{2}$  *equals* the distinguished lepton transport value  $\delta = \frac{1}{2}$  that v20 used by hand — so the geometry plausibly *supplies*  $\delta$  (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g.  $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$ ) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the  $\delta = \frac{1}{2}$  resolvent (closed combinatorially via the word-length  $R$ ); the holonomy supplies  $\delta$  and the leg selection, not the amplitude magnitude.  $c_u/c_d$  is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ( $\delta = \frac{1}{2}$ ). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41\_leg\_assignment.py]

**The torsion normal form and the  $\delta = \frac{1}{2}$  theorem ([E]).** The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness =  $\mu_4$  torsion [E]:* for any  $M$  (symbolic),  $\prod_k U^k M U^{-k} = (MU)^4$  — the  $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$  is a fourth root of unity”: the  $\mu_4$  atom appears as the *order* of the twisted cusp generator. *(ii) The  $\delta = \frac{1}{2}$  theorem [E] (both directions):* on the *involutive* branch ( $T^2 = \mathbf{1}$ )  $T$  is a Hermitian reflection  $2vv^\dagger - \mathbf{1}$ , and the cusp trace condition  $2v^\dagger U^{-1}v = 1$  splits *exactly* into  $|v_1|^2 = \frac{1}{2}$  (real part) and  $|v_2| = |v_3|$  (imaginary part), forcing

$$\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}$$

( $\chi_M = \lambda^3 - 1$  from unitary + trace 0 + det 1). So  $\delta = |M_{22}| = \frac{1}{2}$  holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a  $\mu_4$ -torsion *identity*, not an observation. *(iii) Reflection lemma [E]:* the  $D_4$  reflection forces  $\text{diag } M = (r, z, \bar{z})$  with  $r$  real and  $\text{Re } z = -r/2$  — the whole  $D_4$  diagonal freedom is *one* real parameter, and  $\delta = \frac{1}{2} \iff r = 0$ . *(iv) Branch census [E] (seeded):* the full  $D_4$ -fixed flat cusp locus realises only  $\text{tr}(MU) \in \{1, -1\}$ ; the involutive branch has the diagonal *constant*  $(0, \frac{1}{2}, \frac{1}{2})$  to machine precision, while on the  $\{1, i, -i\}$  branch  $d_1$  varies over  $[0, 1]$  — and the explicit v33/v40 point sits exactly in its  $d_1 = 0$  slice, the *same*  $\delta$  value. *Residue [C] (recorded):* whether the anchor splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (Mehta–Seshadri side) forces the  $d_1 = 0$  slice — or selects the involutive branch — is the remaining question;  $\delta = \frac{1}{2}$  itself is no longer the open item. [verification/v114\_torsion\_delta.py]

**The anchor pins the residue: an exact normal form for the ( $U_{\text{wall}}$ ) bundle ([E]).** The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an

*exact closed form.* (i)  $\mu_4$ -average lemma [E]:  $\sum_k U^k X U^{-k} = 4 \operatorname{diag} X$  for any  $X$  —  $\mu_4$  conjugation-averaging is the diagonal projection, so the exponents at infinity are  $4 \operatorname{diag} A_0$  and the anchor splitting holds iff  $\operatorname{diag} A_0 = (2, 1, 1)/4 = a/|\mu_4|$ : the residue diagonal is the anchor over  $\mu_4$  (v33's diagonal was forced, not chosen). (ii) The  $(8, 0, 5)/144$  lemma [E]: the anchor diagonal and the cusp spectrum  $\{0, 1, 2\}/N_{\text{fam}}$  pin  $\sum |a_{ij}|^2 = \frac{13}{144}$ , and with  $a_{13} = 0$  the determinant condition solves uniquely to

$$\boxed{(|a_{12}|^2, |a_{13}|^2, |a_{23}|^2) = \frac{(8, 0, 5)}{144}} \quad \begin{array}{l} 8 = \operatorname{rank} E_8, \quad 5 = g_{\text{car}}, \\ 144 = (|\mu_4| N_{\text{fam}})^2, \quad 8+5=13=\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same*  $\Delta_Q$  that denominates  $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$ . (iii) *Exact normal form* [E]:  $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$  has  $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$  exactly. (iv)  $A_0^*$  is the RH solution [E]: its big-circle monodromy is trivial to  $10^{-10}$ ,  $\prod M_k = \mathbf{1}$ , and the unitarised holonomy has  $|\operatorname{diag} \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ ,  $\operatorname{tr}(\widetilde{M}_0 U) = 1$  — the v33/v40 point is gauge-equivalent to  $A_0^*$  (the  $a_{12}, a_{23}$  phases are diagonal gauge). (v) *The anchor forces the  $d_1 = 0$  slice* [E]: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with  $|\operatorname{diag} \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$  — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**;  $\delta = \frac{1}{2}$  needs no selection. *Residue* [C]:  $a_{13} = 0$  and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining  $U_{\text{point}}$  step. [verification/v115\_anchor\_residue.py]

**The resonance theorem:**  $M_\infty = \mathbf{1} \iff a_{13} = 0$  — **one gauge orbit** [E]  
([verification/v116\_resonance\_uniqueness.py])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as  $dY/dw = -(1/w)(B_0 + B_1 w + \dots)Y$  with  $B_m = \sum_k i^{km} U^k A_0 U^{-k}$ , the *twisted*  $\mu_4$  averages collapse [E]:

$$B_0 = 4 \operatorname{diag} A_0, \quad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio  $-i$  cells survive). The anchor exponents  $\operatorname{diag}(2, 1, 1)$  have resonance gap 1; the level-1 formal-gauge equation  $(1 - \operatorname{ad} R_0)H_1 = R_1$  is singular exactly on the cells  $\{(2, 1), (3, 1)\}$ , where  $B_1$  has entries  $(0, 4\overline{a_{13}})$ ; all higher levels are non-resonant and the formal monodromy is  $e^{-2\pi i \operatorname{diag}(2, 1, 1)} = \mathbf{1}$ . Hence [E]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary* [E]:  $a_{13} = 0$  + the  $(8, 0, 5)/144$  moduli + phases = diagonal gauge  $\Rightarrow$  the flat anchor locus is **exactly one gauge orbit**, the exact  $A_0^*$ : within the  $\mu_4$ -equivariant anchor class the  $(U_{\text{wall}})$  bundle datum is *algebraically unique*. *Sharpness* [E]:  $\|M_\infty - \mathbf{1}\| \sim 8.5|a_{13}|$  (0.17 at  $a_{13} = 0.02$ ) vs  $10^{-10}$  at  $A_0^*$ . *Honest scope* [C]: the equivariant ansatz is the established  $D_4$  selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal  $(0, \frac{1}{2}, \frac{1}{2})$ ) remain [E] — transcendental monodromy of an exact system.

**The monodromy is the Weyl group of  $A_3$  — the holonomy is exact** [E]  
([verification/v117\_monodromy\_weyl\_a3.py])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact  $A_0^*$  system is itself *exact*, with entries in  $\frac{1}{2}\mathbb{Z}[i]$ :

$$\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}$$

unitary,  $\det = 1$ ,  $\operatorname{tr} = 0$ ,  $\chi = \lambda^3 - 1$  (the cusp class *exactly*),  $\widetilde{M}_0^3 = \mathbf{1}$ , and  $\operatorname{diag} \widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$  — so  $d_1 = 0$  and  $\delta = \frac{1}{2}$  are now *matrix entries*, not observations;  $(\widetilde{M}_0 U)^4 = \mathbf{1}$  with  $\operatorname{tr}(\widetilde{M}_0 U) = 1$  (the  $\{1, i, -i\}$  branch,  $d_1 = 0$  slice — exactly where v114/v115 located the physical point). *The group* [E]: exact

enumeration of  $\langle U, \widetilde{M}_0 \rangle$  gives order 24 with order statistics (1, 9, 8, 6) and character values (3, -1, 0, 1) — uniquely  $S_4$  in its standard  $\otimes$  sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice**  $A_3$ :

$$\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \quad |W(A_3)| = 4! = 24$$

— the flavor wall realises the  $A_3$  factor of the glue  $D_5 \oplus A_3 + \mu_4$  as its Weyl group ( $U$  = a 4-cycle, the  $\mu_4$  deck;  $\widetilde{M}_0$  = a 3-cycle, the family rotation). *Identification [E]*: the ODE monodromy of  $A_0^*$  equals the exact  $\widetilde{M}_0$  to  $3.5 \times 10^{-10}$ , and the invariant form  $H$  is  $U$ -invariant to  $10^{-11}$ . *Status*: holonomy values [E]; only the analytic identification stays [E]. The  $\delta$  thread (v20  $\rightarrow$  v40/v41  $\rightarrow$  v114  $\rightarrow$  v115  $\rightarrow$  v117) is closed end to end.

**The hexagon is the sign-twisted family spectrum — first exact H2 piece [E]**  
( [verification/v118\_hexagon\_family\_dictionary.py] )

The missing “ $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary” (H2) acquires its first exact entry. *Sign-twist lemma [E]*:  $-\widetilde{M}_0$  has order 6 and

$$\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ( $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$  realised as  $(-1) \times \langle \widetilde{M}_0 \rangle$ ); the v20 denominators  $\frac{5}{4} - \cos(r\pi/3)$  are exactly the eigen-denominators  $|1 - \zeta_6^r/2|^2$  of the two family resolvents  $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$ . *Cyclotomic determinant [E]*:  $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1 - t)\Phi_3(t)$ , so  $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$  — and the lepton coefficients become resolvent determinants:  $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$ ,  $c_\mu = N_{\text{fam}}/(2\det_+) = \frac{4}{3}$ ,  $c_\tau = |\mu_4|\det_-/N_{\text{fam}} = \frac{7}{6}$ , product rule  $= |\mu_4|/\det_+ = \frac{32}{9}$ . The lepton 7 and 9 *are* the family-resolvent determinants;  $\delta = \frac{1}{2} = |\widetilde{M}_0|_{22}$  is supplied by the matrix itself. *Sheet-extended group [E] (audit)*:  $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$  has order  $48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$  — the seed’s quartic coefficient. *Residue [C]*: the dictionary’s *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20’s established input / open — nothing re-fished.

**The address table: the lepton words are the compiler atoms [E]/[C]**  
( [verification/v120\_address\_table.py] )

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today’s readings (the red-team firewall). (i) *The lepton words are the atoms [E]*:

$$(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}}) = (p_0 + e_2, e_2, p_0)$$

in mass order (longest word = lightest lepton). (ii) *Addresses = division by the hexagon [E]*:  $(r, w) = L \bmod p_2$  with  $p_2 = 6 = |R^+(A_3)|$  itself an atom:  $e \rightarrow (2, 1)$ ,  $\mu \rightarrow (5, 0)$ ,  $\tau \rightarrow (3, 0)$ ;  $\sum r = 10 = p_3 = A_\Lambda$ ,  $\sum L = 16 = \dim S^+$ . (iii) *Sheet parity [E]*: by the v118 dictionary,  $e$  sits on the *untwisted* sheet ( $\omega \in \text{spec } \widetilde{M}_0$ ),  $\mu, \tau$  on the *twisted* sheet  $(-\omega, -1)$ ; the anchor lepton  $\tau$  occupies the unique *real* twisted eigenvalue  $-1$  — exactly the site where v20’s product rule replaces the resolvent. (iv) *Quark sum rules [E] (frozen v18 words 7, 7, 5, 3, 2, 0)*: up-type sum  $= 10 = p_3$ ; down-type sum  $= 14 = p_1 + p_3$ ; **all quarks**  $= 24 = |W(A_3)|$  (the monodromy group order, v117); **all nine charged fermions**  $= 40 = p_1 p_3 = a^T R a$  (v119); quark  $\sum r = 2p_2$ ,  $\sum w = |\mathbb{Z}_2|$ ; the top sits at the vacuum site  $(0, 0)$  — zero word, no suppression, the established ladder anchor. *Residue [C]*: five independent closures (16, 10, 14, 24, 40) constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

**The address table is pinned:  $R$  is the unique hexagon matrix with atom margins [E]**  
( [verification/v121\_address\_pinning.py] )

The per-fermion assignment closes at the *information* level. The established identity  $L = R + 2U$  (v95) says the word-length table *is*  $L[\text{sector}][\text{generation}]$  — rows (up; down; lepton)  $= ((7, 3, 0); (7, 5, 2); (8, 5, 3))$ , the v18/v20 words verbatim — so the addresses are the entries of the *residue operator*  $R$  plus one hexagon turn for the first generation, and  $R$ ’s first column is the **anchor**  $a = (1, 1, 2)$ . The margins are atoms [E]: row sums  $(4, 8, 10) = (e_1, \text{rank } E_8, p_3)$ , column sums  $(4, 13, 5) = (e_1, \Delta_Q, g_{\text{car}})$ . And the

census closes it [E]:

among all  $3 \times 3$  matrices with entries in the hexagon  $\{0..5\}$ ,  
exactly 17 have the atom margins — and exactly ONE has  $\det = 8$

— and it is  $R$  (also the unique one with SNF = (1, 1, 8); all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ( $m \sim \lambda_Y^L$ : longer word = lighter fermion). *Residue [C]*: the open H2 item shrinks from “derive 9 per-fermion integers” to “derive the six atom margins +  $\det = \text{rank}$ ”. *Firewall*:  $R$ ,  $L$  and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

**The margins are theorems** ([verification/v122\_margin\_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — (S1) the  $D_4$  annihilator  $n = (5, -9, 6)$  with  $n^T R = (8, 0, 0)$  (it kills the unchanged  $(c_2, c_3)$  plane;  $n \cdot a = 8$ ), (S2)  $\det R = 8$ , (S3)  $\det K = 4$  with  $K = R + Q\Sigma$  — pin  $R$  *uniquely* among hexagon matrices [E]: the (S1) census has  $4 \times 4 \times 4 = 64$  candidates, (S2) leaves 12, (S3) leaves **exactly one**, and it is  $R$ . Hence the atom margins, the anchor column  $Re_1 = a$  and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]*: on the (S1) + (S2) locus  $\det(M + 2U) = 20 = \det L$  holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler’s familiar overdetermination. *Residue [C]*: H2 reduces to the establishment of the selectors themselves ( $n$  and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

**The dual-normal selector:**  $(d, n)$  **pins**  $R$  **columnwise** ([verification/v136\_dual\_normal\_selector.py]). The selector list shrinks once more. With the dual anchor  $d = a^\top R^{-1}$  ([verification/v134\_dual\_anchor.py]) and the torsion normal  $n$ , each *column* of  $R$  obeys  $d \cdot c_j = a_j$  and  $n \cdot c_j = 8\delta_{1j}$ ; the common kernel of  $(d, n)$  is the primitive vector  $(6, 8, 7)$  (audit:  $p_2$ , rank  $E_8$ , scalaron-7), so within the address box  $\{0..5\}^3$  each column is the *unique* lattice point (census: 1 of 216 each) —  $(d, n) \Rightarrow R$ , and the  $6^9$  census above collapses to three column problems [E]. *Honesty*:  $K$  is *not* pinned directly ( $d^\top K = (1, \frac{1}{2}, 1)$ );  $\det K = 4$  comes via the  $\mu_4$  wall ([verification/v135\_det\_surface.py]). *The mechanism is bounded*: the zero mode of  $A_0^*$  carries the exact spectral normal  $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$  (signed squares;  $\|v\|^2 = \frac{16}{9}$  with  $16 = \dim S^+$ ,  $\text{adj } A_0^* = \frac{2}{9}P_v$  with  $\frac{2}{9}$  = the SdS entropy curvature),  $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$ ,  $\sigma \cdot a = N_{\text{fam}}$ ,  $n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$ ; and the exact  $W(A_3)$  orbit control shows *no* group image of  $\sigma$  is proportional to  $n$  — the naive cofactor/orbit lift fails, the step  $\sigma \rightarrow n$  is a genuine mechanism beyond the monodromy group.  $R4'$  restated [C]: derive the *pair*  $(d, n)$ ; then  $R$  is forced columnwise.

**The selector triangle** ([verification/v139\_selector\_triangle.py]). The pair itself now collapses further. First,  $d$  is *pure anchor data*:  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  exactly — the closed form never mentions  $R$ , so the first selector is *derived*, not residual. Second, the frame  $(\mathbf{1}, a, \sigma)$  is a basis of  $\mathbb{Z}^3$  with  $\det(\mathbf{1}|a|\sigma) = 11 = \|\text{Pl}(K)\|_1$  (the quark constant as the frame volume), and  $n$  is the *unique* integer covector with the three atom pairings

$$n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \quad n \cdot a = 8 = \text{rank } E_8, \quad n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$$

— the cofactor definition is replaced by frame data plus three atoms. On the  $(\mathbf{1}, a)$ -constrained line the  $\sigma$ -pairing is  $11(11+t)$ : divisible by 11 for *every*  $t$ , a perfect square exactly at the true  $n$  ( $t=0$ ). The review’s predicted lift is exact as an audit identity:  $n = \text{reverse}(\sigma) + |\mu_4|e_3$  (generation flip +  $\mu_4$  lift at the third slot).

**Frame integrality: three pairings  $\rightarrow$  two** ([verification/v142\_frame\_integrality.py]). The third pairing is *not* independent. The pairing triples  $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$  of *integer* covectors form an index-11 sublattice of  $\mathbb{Z}^3$ , cut out by the single congruence

$$x - 3y + z \equiv 0 \pmod{11} \quad (\text{index } 11 = \|\text{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8) = (2, 8)$ , integrality *forces*  $z \equiv 0 \pmod{11}$  — the  $11(11+t)$  line *is* the integer line. Primitivity forces  $t$  *even* (odd  $t$  makes every entry of  $n(t)$  even), and  $t=0$  is the unique even  $t$  with a square  $\sigma$ -pairing for  $|t| < 88$  (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities



( $n \cdot a = \det R$  by Laplace along the anchor column;  $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$  with  $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ , v134) — restructuring, not establishment.

**The pairing values are atom identities** ([verification/v145\_pairing\_atoms.py]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally —  $n = w_0(\sigma) + |\mu_4|e_3$  with  $w_0$  the  $A_3$  *exponent duality*  $m \leftrightarrow h-m$  (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n \cdot \mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \quad n \cdot a = |\mu_4| a_3 = 8 = \text{rank } E_8 \text{ via } |\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9, \sigma \perp w_0(a))$$

and  $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4|g_{\text{car}} = 110 - 9 + 20 = 121$  with  $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2|g_{\text{car}} \|\text{Pl}(K)\|_1$ .  $R_4'$  *residue* [C] (*final form*): derive the *one* map  $n = w_0(\sigma) + |\mu_4|e_3$  — no independent number remains; everything else in the chain anchor  $\rightarrow (d, n) \rightarrow R$  is exact.

**Both dual normals are anchor data** ([verification/v149\_cusp\_normal.py]). The remaining opacity dissolves in the canonical frame: the pairings of  $n$  with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$n \cdot e_1 = 6 = p_2(a), \quad n \cdot e_2 = 3 = p_0(a), \quad n \cdot e_3 = 5 = e_2(a)$$

— one anchor invariant per  $Q_+$  degree, and the cusp matrix is unimodular, so these three pairings determine  $n$  uniquely. With  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data (6, 3, 5), the frame pairings (2, 8, 121) and the  $w_0$ -lift are three presentations of one covector, all atomic (equivalences exact). *Audit*:  $\sigma \rightarrow (5, 1, 2)$ ; alternate reading (6, 3, 5) =  $(|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$ ; sum 14 =  $\dim G_2$ .  $R_4'$  *residue* [C] (*cusp form*): derive the *assignment* — why the  $Q_+$  degrees (1, 2, 3) carry  $(p_2, p_0, e_2)$  in that order; one discrete assignment, zero continuous freedom.

### Exterior Leg Lemma — the quark $u/d$ leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end.  $u$  and  $d$  share the hexagon-radial data  $(r, w) = (1, 1)$  ( $L = 7$  for both), and the  $C_6$  resolvent  $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$  reads *only*  $(r, w)$  — so any *scalar* leg gives  $c_u/c_d \equiv 1$ . A scalar sees radius, not orientation. The  $u/d$  separation lives in the smallest *non-scalar* invariant of the  $SU(3)_F$  composition: the exterior square  $\Lambda^2 F$ , i.e. the oriented anchor-plane area  $\text{Pl}_{1,a}(K)$ . The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \begin{pmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{pmatrix} \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

**Typing:** [E] for the Plücker / microcode identities ( $\text{Pl}(K) = (-1, 6, 4)$ ,  $\|\cdot\|_1 = 11$ ,  $5 \cdot 11 / (9 \cdot 13) = 55/117$ ); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio  $\mathcal{C}_{ud}(\rho^*)$  = this  $\Lambda^2 F$  readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on  $\Lambda^2 F$ . The quark residual is thereby re-typed from “find the right scalar  $y$ ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale ( $U_{\text{point}}$ ) stays as an anchor, no new scalar, no fishing.

*Audit* (the 121 lemma, [verification/v119\_review\_validation\_2.py]): the “11” has a second, independent appearance [E]: with  $h = \mathbf{1} + a = (2, 2, 3)$  in the canonical anchor ordering,  $h^T R h = 121 = 11^2 = \|\text{Pl}(K)\|_1^2 = (p_3 + 1)^2$  — the quadratic *anchor norm* of the residue operator  $R$  reproduces the Plücker integer (sensitivity control: the three orderings of  $h$  give {105, 121, 135}; only the canonical one yields 121). Bonus checksums:  $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$  (the entry sum of  $R$ ) and  $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$ . Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42\_exterior\_leg.py]

**Bridge test** ([verification/v43\_exterior\_bridge.py]). The typing is confirmed: for the  $SU(3)$  holonomy the exterior square is the conjugate representation,  $\Lambda^2(\widetilde{M}) = \widetilde{M}$  exactly — the exterior leg *is* the  $\bar{\mathbf{3}}$ . But the *continuous* action  $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$  does *not* equal the *integer*  $\text{Pl}(K) = (-1, 6, 4)$ : the integer Plücker is the combinatorial  $K$  datum. So the bridge  $\rho^* \rightarrow \text{Pl}(K)$  is the *discrete* non-abelian-Hodge invariant (the integer  $R$ ;  $\det R = 8$  and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the  $\Lambda^2 F$  side.

**Lie-level grounding — no special math** ([verification/v44\_carrier\_exterior.py]). The discrete

invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier,  $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$ , and under  $SU(5) \subset SO(10)$  the up quark sits in  $10 = \Lambda^2(5)$  while the down sits in  $\bar{5} = \Lambda^4(5)$ . So the *exterior degrees* are  $\deg(u^c) = 2$ ,  $\deg(d^c) = 4$ , with  $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$  (the hexagon) and  $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$  (the sheet). The “ $\Lambda^2$ ” of the exterior leg is therefore not exotic Hodge data — it is the carrier’s own  $\Lambda^2$  grading (the up quark *lives* in  $\Lambda^2(5)$ ). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [E] fact. *Honest scope*: this grounds the *type*; the Plücker *number* 55/117 stays the family  $\text{Pl}(K)$  [E], and the carrier- $\Lambda^2 \leftrightarrow$  family- $\Lambda^2 F$  link (via family  $\otimes$  carrier = 15 = 16–1) plus the normalisation remain [C].

**The “11” is Pascal — a third, simplest origin** ([verification/v45\_family\_exterior.py]). Following that link,  $\text{Pl}(K)$  is the *exterior algebra of the family fundamental*  $4 = \mu_4$  (the  $A_3 = SU(4)$  fundamental):  $\Lambda^k(4) = \binom{4}{k}$  is Pascal row 4 = (1, 4, 6, 4, 1) with full sum  $2^4 = 16 = \dim S^+$ , and  $|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$  are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative  $\Lambda^\bullet(\mu_4)$  up to degree 2 =  $\deg(u^c)$ ), and family  $\otimes$  carrier = 15 =  $\dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$  read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

**Lemma 6** (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that  $c_u/c_d = C_{ud}(\rho^*)$  must be evaluated at the uniquely selected  $\rho^*$  without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. **This is now resolved (v49):**  $C_{ud}$  is constant on the discrete selector stratum  $S_{8,Q}$ , so the “11” is earned as the rigid Plücker readout  $\|\text{Pl}(K)\|_1$  (v42) — independently of point-uniqueness. Point-uniqueness of  $\rho^*$  enters only the absolute amplitude scale, not the ratio.*

**Lemma 7** (U6 — the four certificates).  $C_U^{(1)}$  wall enumeration (done);  $C_U^{(2)}$   $D_4$ -fixed character solver;  $C_U^{(3)}$  selector uniqueness (one  $S$ -equivalence class);  $C_U^{(4)}$  amplitude extraction — the ratios  $R, Q, c_u/c_d$  are read off the (derived) selector stratum (closed, v49/v71); only the absolute  $U_f^*, c_q$  normalisation is the anchor.

**Progress: an explicit valid flat bundle is realised (RH solve) ([E]).** A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue  $A_0$  (eigenvalues  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ ) for which the Fuchsian system  $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$  simultaneously satisfies: the cusp class at each puncture; the splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (exponents  $\sum_k A_k = \text{diag}(2, 1, 1)$ ); the *path-ordered* monodromy at infinity is trivial,  $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$ , i.e.  $M_1 M_2 M_3 M_4 = \mathbf{1}$ ; and the representation is *irreducible* — **case A** (non-trivial  $SU(3)_F$  mixing, not the diagonal case B). So a valid ( $U_{\text{wall}}$ ) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique  $\nabla_F^*$  still needs the  $\det R = 8$  selector and  $c_u/c_d$  still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary (H2). [verification/v33\_explicit\_flat\_bundle.py]

*H2 bridge probed* ([verification/v34\_h2\_bridge\_attempt.py]). The explicit per-puncture monodromies  $M_k$  (cusp class,  $\prod M_k = \mathbf{1}$  to  $10^{-5}$ ) were computed; but the natural resolvent-dressed diagonal extraction ( $|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$  times the  $C_6$  resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise  $\Gamma_{ij}^{\min}$  geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family  $\rho^*$ ) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the  $W(A_3)$  monodromy, not the diagonal dressing — the probe’s failure is thereby explained, and the dictionary is exact.*

**Theorem U — the four-way split (sharper than one monster gate) [E]/[O]**

$D_4$  fixes the admissible chamber, not the physical point (the  $D_4$ -fixed locus is positive-dimensional, U3); the discrete selectors  $\det R = 8$ ,  $\text{SNF}(R) = (1, 1, 8)$ ,  $\text{Spec}(Q_+) = \{1, 2, 3\}$  do the cutting. The gate splits into four pieces, three of them essentially closed:

- $U_{\text{unitary}}$  [E]: parabolic degree 0 + polystable  $\Rightarrow$  unitary (Mehta–Seshadri)  $\Rightarrow \Phi = 0$ ; the harmonic metric is the *finite* invariant form  $H$  ( $M_k^\dagger H M_k = H$ ), not a Hitchin PDE ([verification/v40\_harmonic\_metric.py]).
- $U_{H_2}$  [E]:  $R, Q, K$  are read off the bundle (splitting type =  $a$ ,  $\det R = 8$ ,  $\text{Spec}(Q_+)$ ) ([verification/v39\_uwall\_selectors.py]).
- $U_{\Lambda^2}$  [E]: **Readout Rigidity** — on the discrete stratum  $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$  the anchor-plane readout is *constant*,  $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$ , *independent* of the continuous  $D_4$  position — so  $c_u/c_d$  needs no point uniqueness ([verification/v49\_readout\_rigidity.py], [verification/v42\_exterior\_leg.py]).
- $U_{\text{point}}$  [O]: full uniqueness of  $\rho^*$  — needed only for the *complete*  $U_f^*$  amplitude matrix, *not* for  $c_u/c_d$ .

**Headline:** the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only  $U_{\text{point}}$  stays open.

**The simple closure** ([verification/v71\_simple\_r\_bridge.py]). The selector stratum  $\mathcal{S}_{8,Q}$  is now *fully derived*:  $\det R = 8 = \text{rank } E_8$  with  $\text{SNF}(R) = (1, 1, 8)$  is the lattice invariant (cf.  $\det Q = 3 = N_{\text{fam}}$ , v70), and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  is the  $D_4$ -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants ( $\det Q, \det K, \det R, \det L$ ) = (3, 4, 8, 20), product  $1920 = |W(D_5)|$ . So by Readout Rigidity the quark *ratios* ( $c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$ , etc.) are *pure integer Plücker readouts* — *no transcendental monodromy solve*. The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece,  $U_{\text{point}}$ , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensional scale, not a missing computation.

**Gate 1 complete:**  $U_{\text{point}} \rightarrow v_{\text{geo}}$  ([verification/v75\_upoint\_to\_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ( $c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , v20), every cross-sector  $c$ -ratio is closed (Plücker), and each *sector product* is a clean  $\varphi_0^{\text{ret}}$ -power (Grand Mass Volume, v46:  $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$ ; the lepton product is  $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ ). Since (pairwise ratios) + (product)  $\Leftrightarrow$  (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale*  $v_{\text{geo}}$ . **So  $U_{\text{point}}$  is not an independent gate: it reduces to  $v_{\text{geo}}$ , the same one irreducible dimensionful anchor as gravity's  $1/G$  (v68) — the two [O] anchors collapse to one.**

**Research Contract 2 — ( $G_{\text{metric}}$ )**

**Goal.** Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed  $R + R^2$  equation.

**Lemma 8** (G0–G1 — configuration space & BRST/Hodge Fredholm). *For  $s > 5/2$  a Sobolev space of  $\Theta$ -symmetric metrics on the doubled seam collar  $M_\Sigma$  admits a gauge slice; the gauge-fixed gravitational complex  $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{K^*} \Omega^0(TM)$  with  $\Delta_{\text{gh}} = K^* K$  and  $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}} \Pi_{TT}$  is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct-/Cadabra; linear-algebra certificates in Lean (medium).*

**Lemma 9** (G2 — finite relative spectral action, *heat-kernel grounded* [E]).  $S_{\text{rel},\chi}$  is finite, differentiable, with a local Seeley–DeWitt expansion  $\text{Tr} f(D^2/\chi^2) \sim f_4 \chi^4 a_0 + f_2 \chi^2 a_2 + f_0 a_4 + \dots$ . For

the Lichnerowicz Dirac operator  $D^2 = \nabla^2 + \frac{R}{4}$  (so  $E = -\frac{R}{4}$ , spinor trace  $\text{tr } \mathbf{1} = 4$ ) the Gilkey coefficients give, per  $(4\pi)^{-2}$ ,

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein-Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the  $R^2$ /Starobinsky term), so the spectral action structurally produces  $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}^2}) + \dots$  with the scalaron mass ratio fixed by the cutoff moment,  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ . The TFPT closure  $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$  fixes  $f_0 = 6(4\pi)^2/c_3^7$ , i.e.  $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$  — the boundary normalisation  $c_3$  is the gravitational scale — and hence the Starobinsky slow-roll forms  $n_s \simeq 1 - \frac{2}{N}$ ,  $r \simeq \frac{12}{N^2}$ . The  $R + R^2$  structure is convention-independent (standard Gilkey); the precise rational  $\frac{1}{72}$  and factor 6 are the standard Dirac conventions used here ([[verification/v36\\_spectral\\_action\\_g2.py](#)], [[verification/v28\\_gravity\\_fR.py](#)]).

**The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector ([E]).** The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5):  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a *strictly positive* effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| admissible IR sector = closed under stated RP;<br>ambient metric measure (G6) = $G_{\text{metric}}$ gate |
|----------------------------------------------------------------------------------------------------------|

G2 supplies the local action ( $R + R^2$ , heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ( $G_{\text{metric}}$ ) to the ambient projective limit (G6). **Precise status (Alessandro’s boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target*. Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it “mere mathematical completeness”. [[verification/v36\\_spectral\\_action\\_g2.py](#)]

**The glue Q-system: the index-4 object of the  $G_{\text{net}}$  statement exists explicitly ([E]/[C]).** The  $G_{\text{net}}$  target — “the seam-Calderón inclusion has Jones index  $4 = |\mu_4|$ ” — no longer refers to an unknown object. On the glue group algebra  $A = \mathbb{C}[\mathbb{Z}_4]$  (multiplication  $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$ , unit  $\delta_0$ ) all Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity  $(m \otimes 1)(1 \otimes m^*) = m^*m = (1 \otimes m)(m^* \otimes 1)$ , and specialness  $mm^* = |\mathbb{Z}_4| \mathbf{1}$  — a special  $C^*$  Frobenius algebra with

|                                       |
|---------------------------------------|
| $\dim \theta = 4 =  \mu_4  = [B : A]$ |
|---------------------------------------|

exactly the Carrier Index Lemma value (KLM:  $\mu_{\text{carrier}} = 16 = 4^2 \mu_{E_8}$  re-verified). *Locality = isotropy* [E]: on the Lagrangian glue  $\langle (1, 1) \rangle$  the discriminant form gives  $q(k(1, 1)) = k^2 \in \mathbb{Z}$  for every  $k$  — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step* [E]:  $\{0, 2\} < \mathbb{Z}_4$  closes and gives the unique intermediate Q-system  $\mathbb{C}[\mathbb{Z}_2]$  = the  $SO(16)_1$  tower step, with index factorisation  $4 = 2 \times 2$ . *R2 sharpened* [C] (recorded, not claimed): what remains is the operator-algebraic *identification* “the seam-Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system” — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

**The hull is the graded module of the Q-system** ([[verification/v128\\_graded\\_hull.py](#)]). The Q-system is not merely an abstract companion of  $E_8$  —  $E_8$  *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in  $D_5 \oplus A_3$ +glue coordinates, decompose as  $240 = 52 + 64 + 60 + 64$  over



the glue  $\mathbb{Z}_4$  (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root,  $\text{coset}(r_1 + r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$ :

$E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier; the grading group is the glue = the Q-system

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that  $E_8$  already *carries*.  
[verification/v125\_glue\_qsystem.py]

**Graded Frobenius** ([verification/v143\_graded\_frobenius.py]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside  $E_8$  [E]: (i) *coset duality* —  $\text{coset}(-r) = -\text{coset}(r) \bmod 4$  for all 240 roots ( $-C_1 = C_3$  bijectively,  $C_0, C_2$  self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing  $g_k \times g_{-k} \rightarrow \mathbb{C}$ ; (ii) the glue character average  $\frac{1}{4} \sum_k i^k \deg$  is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index  $|\mathbb{Z}_4| = 4$ ); (iii) each glue sector is *one*  $W(D_5) \times W(A_3)$  orbit with multiplicity-one dimensions  $64 = 16 \cdot 4 = (\mathbf{16}, \mathbf{4})$ ,  $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$  — simple currents, sector ring  $= \mathbb{Z}_4 =$  the  $\mathbb{C}[\mathbb{Z}_4]$  fusion, halfway subgroup  $\{0, 2\} = SO(16)_1$ . *R2 residue* [C] (*final form*): exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

**Where that statement lives: the NS/R sector census** ([verification/v148\_fock\_census.py]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* ( $D_5, SO(6)$ ) weights, and integer weights lie in the even discriminant classes — so the NS module supports only  $\{(0, 0), (2, 0), (0, 2), (2, 2)\}$ : **the odd glue classes  $k(1, 1)$  — the actual  $\mu_4$  simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ( $2^8 = 256$  ground states) carries exactly the four odd class pairs, 64 each, and splits  $128 + 128$  into the odd sectors of the *two* Lagrangian glues  $\langle(1, 1)\rangle$  and  $\langle(1, 3)\rangle$  (v92): *choosing the glue is choosing the Ramond projection* ( $128 =$  one  $SO(16)$  spinor chirality), and  $E_8$  assembles as  $248 = 120_{\text{NS}} + 128_{\text{R}}$ . *Consequence* [C]: the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

**Lemma 10** (G3–G4 — reflection positivity, fixed and integrated). *For each  $\Theta$ -invariant  $g$  in the Calderón ball the physical boundary kernel  $K_g = P_{\text{adm}} \Pi_{\text{phys}} \mathcal{N}_g^{TT} \Pi_{\text{phys}} P_{\text{adm}}$  is reflection positive,  $\langle \Theta F, F \rangle_{K_g} \geq 0$  (G3); and  $d\mu_{\text{Cal}, \chi}$  is  $\Theta$ -invariant, supported where G3 holds uniformly, so  $\int \langle \Theta F, F \rangle_{K_g} d\mu_{\text{Cal}, \chi} \geq 0$  (G4). **G4 is the decisive step: RP after integrating over dynamical metrics, not merely at fixed background.***

**Lemma 11** (G5 — gap dominance [E]).  $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$ , hence  $2\|V\| = 0.7852 < 2.4328$  and  $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$ . *The numerical inequality is trivial* ([verification/v29\_research\_contract\_certs.py]); *the work is the operator-norm bound (hard).*

**Sugawara reading + atomic OS moment** ([verification/v171\_os\_moment\_cluster.py]). The G5 budget is the  $E_8$  Sugawara constant:  $248c_3^2 = \frac{31}{8\pi^2}$  with  $31 = 1 + h^\vee(E_8)$  and  $c((E_8)_1) = 248/31 = 8$ , so gap dominance reads  $|R^+(A_3)| \log(N_{\text{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$  — the  $A_3$  transfer gap beats the  $E_8$  Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*:  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  makes every two-point correlator  $C(n) = A + B(64/729)^n + C(1/729)^n$  the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant  $729/665$ , correlation length  $\xi = 1/(6 \log \frac{3}{2}) = 0.4111$ ).

**Theorem 1** (Decoupling theorem — physical low-energy claims do *not* need the full  $G_{\text{metric}}$ ). *If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,*

$2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$  (G5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap  $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$  protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings,  $\alpha^{-1}$ , the  $R+R^2$  regime) is independent of the open ambient measure. Symbolically

$$\boxed{\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion.}}$$

Thus  $(G_{\text{metric}})$  is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit (G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [E]; the operator-norm bound on  $\|V_{\text{metric}}\|$  is the conditional input.

**Lemma 12** (G6–G7 — projective limit & Ward identities). A projective system  $\mu_{\chi_n}$  with  $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$ , uniform moment bounds  $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$ , tightness and regulator-independence yields  $\mu_{\text{QG}} = \lim \mu_{\chi_n}$  on  $\mathcal{G}/\text{Diff}$  (G6, the hardest step); the limit obeys the BRST Ward identities  $\langle s\mathcal{O} \rangle = 0$ , whence  $\nabla^\mu T_{\mu\nu} = 0$  and the  $\eta$  boundary phase cancels the APS anomalies (G7).

**Gate 2 reduction:** G6 is a boundary projective limit, not a bulk one  
([verification/v76\_gmetric\_reduction.py])

The exact decoupling margin is  $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2} (31 = 2^{g_{\text{car}}} - 1)$ . **Holographic reduction:** because the seam is a finite causal boundary, the ambient bulk metric measure is reconstructed from a finite-codimension seam (Calderón) boundary measure — so G6 is a boundary projective limit. The conditional theorem is then

$$\boxed{\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse “full QG missing”. [C]/[O]

**And that boundary measure is a rigorously constructed net** ([verification/v77\_e8\_conformal\_net.py]). The level-1 central charges  $c = \dim \mathfrak{g}/(1 + h^\vee)$  give  $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$ ,  $c(D_5)_1 = \frac{45}{9} = 5$ ,  $c(A_3)_1 = \frac{15}{5} = 3$ , and  $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$  is a conformal embedding  $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$  (coset  $c = 0$ ). Now  $(E_8)_1$  is the holomorphic  $c = 8$   $E_8$ -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** identify the seam-Calderón boundary measure with the  $(E_8)_1$  lattice net (carrier  $= (D_5)_1 \times (A_3)_1$  subnet), and the gap  $\Delta_{\text{eff}} > 0$  supplies the clustering/tightness — so G6 is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

**The Simple-Current Extension Theorem — the central  $G_{\text{net}}$  closing statement [E]/[C]**  
([verification/v154\_simple\_current\_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension*. Let  $A = (D_5)_1 \otimes (A_3)_1$  be the carrier net and  $L = \langle (1,1) \rangle \subset \text{Disc}(D_5) \oplus \text{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$  the isotropic  $\mu_4$  glue line ( $q(k(1,1)) = k^2 \in \mathbb{Z}$ ). Then the local extension  $B = A \rtimes L$  has

$$\boxed{[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5 + 3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1}$$

so  $B$  is holomorphic (KLM  $\mu$ -additivity), and a holomorphic  $c=8$  chiral net is the lattice net of the unique even-unimodular rank-8 lattice — hence  $B \cong (E_8)_1$  (the same  $c$  rival  $SO(16)_1$ ,  $\mu=4$ , is excluded). Every algebraic ingredient is machine-checked: the  $\mathbb{C}[\mathbb{Z}_4]$  Longo Q-system ([verification/v125\_glue\_qsystem.py]), the  $\mathbb{Z}_4$ -graded Frobenius  $E_8$  hull ([verification/v143\_graded\_frobenius.py]), the Ramond glue sectors ([verification/v148\_fock\_census.py]). **Consequence:**  $G_{\text{net}}$  is internally closed if the seam-Calderón boundary net is defined to be  $A$ ; it stays the one external identification “the RP seam-Calderón

net is  $A$ ” if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not “find a quantum gravity”.

**And that one identification reduces to a single shared premise** ([[verification/v155\\_quasifree\\_boundary.py](#)]). The identification splits (v110) into (a) “the seam Calderón involution is sheet-odd” — the double-cover deck, the same  $|\mathbb{Z}_2|$  that halves  $c_3 = \frac{1}{2 \cdot 4\pi}$  (structural) — and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the bulk covariance, and the compression of a contraction ( $i\Gamma$  with spectrum in  $[-1, 1]$ ) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of  $\Gamma_d$ ). Hence (b) *follows* from “the seam bulk is a reflection-positive free field” — the *same* premise already load-bearing in the area law ([[verification/v59\\_area\\_law\\_evidence.py](#)]) and the EH mechanism ([[verification/v150\\_replica\\_eh\\_model.py](#)]/[[verification/v151\\_bfk\\_split.py](#)]). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by  $G_{\text{net}}$ , the area law and R3, and given it the chain Gaussian bulk  $\Rightarrow$  quasi-free boundary  $\Rightarrow$  one kernel = net (v113)  $\Rightarrow$  carrier  $A \Rightarrow (E_8)_1$  is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

**The construction, exhibited and verified** ([[verification/v156\\_seam\\_net\\_construction.py](#)]). The “ $B \cong (E_8)_1$ ” step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode  $e^{ikx}$  by the harmonic extension  $e^{ikx-|k|y}$ :  $-\partial_n u|_0 = |k|u$ , so for a free bulk the seam operator is the massless chiral dispersion  $\omega_k = |k|$  that defines the free-fermion net on  $S^1$ . Sixteen Majoranas give  $c = 8 = 5+3$ ; the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  (the  $SO(16)$  bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\chi = \frac{E_4}{\eta^8} = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \dots) = j^{1/3}$$

(verified by  $q$ -series), the 248 at level 1 being  $\dim E_8$  — so the net *is*  $(E_8)_1$ . *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

**And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate** ([[verification/v157\\_rigid\\_fixed\\_point.py](#)]). Three facts click together. (i) The DtN/Calderón symbol is *universally*  $|k|$ : the boundary operator of *any* second-order elliptic bulk is an order-1  $\Psi\text{DO}$  with principal symbol  $|\xi|_g$  (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic  $c=8$  theory is *rigid*: every field has weight  $(h, \bar{h}=0)$ , an exactly-marginal deformation needs  $(1, 1)$ , and with  $\bar{h}=0$  *no*  $(1, 1)$  field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same*  $|\mathbb{Z}_2|$  that halves the one-sided  $S^2$  Gauss–Bonnet ( $c_3 = \frac{1}{|\mathbb{Z}_2| 2\pi \chi(S^2)} = \frac{1}{8\pi}$ ) labels the sheet *and* is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ) — one object, three roles, so the geometry fixing  $c_3$  is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6 \log \frac{3}{2} > 0$ ) boundary flow reaches the holomorphic fixed point” — given which, freeness is *forced* by rigidity + uniqueness ( $(E_8)_1$ , v83), not fine-tuned. A strictly smaller and simpler residual.

**And the destination is a proven stable fixed point** ([[verification/v158\\_fixed\\_point\\_stable.py](#)]). An exact operator-dimension count settles it: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty* (bosonic chiral operators have integer  $h$ , lowest 1) — nothing relevant flows the free theory away; the  $h=1$  currents are chiral (v157), not  $(1, 1)$ -marginal; and the lowest genuinely-interacting operator (the quartic,  $h=2>1$ ) is *irrelevant*, as are all higher  $2k$ -fermi operators ( $h=k$ ). So the free chiral  $c=8$  fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ( $\Delta=6 \log \frac{3}{2}$ ) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

**And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise** ([[verification/v160\\_seam\\_gaussianity\\_from\\_pf.py](#)]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  (the signed set-partition/Pfaffian

inversion of  $\langle \psi_1 \cdots \psi_{2n} \rangle = \text{Pf}(\Gamma)$  leaves only the degree-2 kernel — the v113 “one kernel = the net” property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if  $\kappa_{2n}$  were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap  $> 0$  suffices; the value  $(2/3)^6$  only sets the rate). *The honest residual:* v56’s gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — the Schwinger cone the transport must contract has dimension  $\gg 3$ , and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

**And that full-cone premise is now discharged to a single finite one-particle statement** ([verification/v161\_one\_particle\_reduction.py]). The lever is Bogoliubov second quantisation: a quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dimensional Majorana space  $H_\Sigma$ , and the Fock space grades by correlator degree, so  $\Gamma(t)$  acts on the degree- $m$  sector as the exterior power  $\Lambda^m(t)$ . Three exact consequences follow (all machine-checked): (i)  $\text{spec}(\Gamma(t)|_{\text{deg } m})$  is the set of products of  $m$  one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap equals the one-particle gap:** the sub-leading eigenvalue of  $\Gamma(t)$  over *every* degree is  $(2/3)^6$  (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace — the disconnected Wick powers of  $K_\Sigma$  — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation  $\Leftrightarrow$  one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the  $120 = \dim \mathfrak{so}(16)$  weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport’s one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8  $K_\Sigma$  polarisation (v113), sub-leading gap =  $(2/3)^6$  (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of  $(A_2)$ , now with the infinite-dimensional cone eliminated [E]/[O].

**And that finite statement carries no new open content: it factors into the two already-open items** ([verification/v162\_seam\_transport\_identification.py]). Take the one-particle symbol apart into its two data, the gap value ( $\beta$ ) and the fixed space ( $\alpha$ ). ( $\beta$ ) *is forced:* a  $\mu_4$ -equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection,  $\sum_{k=0}^3 U^k X U^{-k} = |\mu_4| \text{diag}(X)$  for  $U = \text{diag}(1, i, -i)$ , and the commutant of  $U$  is exactly the diagonal algebra — so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (the parabolic residue, v115), and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established  $A_3$  subnet grading (v137). ( $\alpha$ ) is the rank-8 Calderón polarisation of the  $\omega_k=|k|$  free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is*  $\mu_4$ -equivariant and admissible — which is the R1 “seam clock” identification (the transfer operator as the seam’s gapped generator) — together with the  $A_2$  net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into  $A_2 + R_1$  with every number forced. The “free-bulk cloud” is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

**And both residual identifications are pinned to their floor, so (A) has no independent open gate left** ([verification/v165\_premise\_a\_closure.py]).  $A_2$  (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net *is*  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally  $|k|$  (v156/v157). So  $A_2$  re-types [O]  $\rightarrow$  [C] (closed by assembly).  $R_1$  (the seam clock) reduces to **GATE.QGEO:** a  $\mu_4$ -equivariant flat gapped generator on the carrier is *unique* ( $= A_0^*$ , v115/v116), so its gap  $(2/3)^6$  is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the  $\mu_4$ -punctured sphere



$\mathbb{P}^1 \setminus \mu_4$  — whose cohomological half is already established (v137,  $b_1=3=N_{\text{fam}}$ ). That is **GATE.QGEO**, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (v153) the irreducibles stay exactly  $\{\pi, v_{\text{geo}}\}$  — closing (A) adds none. **Final accounting:** premise (A) = ( $A_2$  assembly, [C]) + ( $R_1=\text{GATE.QGEO}$ , already open) + the  $\{\pi, v_{\text{geo}}\}$  irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [E]/[O].

**And the two structural residuals — net existence and full-cone reflection positivity — are now discharged to a theorem, leaving the seam realisation as the one irreducible open premise** ([verification/v175\_net\_existence\_full\_cone.py]). The CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the *complete* Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16}=65536$ ); by the exterior-power spectrum theorem every eigenvalue of  $\Gamma(t)$  is a subset product of  $\text{spec}(t)$ , so checking all  $2^{16}$  products shows  $\Gamma(t)$  is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity  $2^8=256$  (the wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading eigenvalue exactly the one-particle gap  $(2/3)^6$  — so full-cone RP/gap reduce *for every*  $m$  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem*. And  $A_2$  is now an *assembled and verified* certificate: the  $(E_8)_1$  character  $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$  (level-1=248), the embedding  $248=120+128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the same input:** that the actual seam Calderón kernel is this gapped one-particle contraction on  $\mathbb{P}^1 \setminus \mu_4$ , i.e. **QGEO.REALIZE.01**, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O] — the single irreducible structural premise; net existence and full-cone RP are [E].

**The No-Unit Theorem —  $v_{\text{geo}}$  is an irreducible metrology primitive [E]/[O]**  
([verification/v153\_no\_unit\_theorem.py])

$v_{\text{geo}}$  is not an open gap. Every load-bearing compiler datum —  $g_{\text{car}}, |\mu_4|, N_{\text{fam}}, \text{rank } E_8, \alpha^{-1}$ , the determinant ladder  $(3, 4, 8, 20)$ ,  $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0^{\text{ret}}$  — is mass-dimension 0 and *invariant* under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass / energy /  $1/G$  carries nonzero dimension. A map from dimension-0 inputs to a dimension- $d \neq 0$  output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** —  $v_{\text{geo}}$  is a *forced input*. The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\text{point}} \sim v_{\text{geo}}, \quad \frac{1}{G} \sim v_{\text{geo}}^2, \quad m/\mu = e^{3/4} \text{ (a pure ratio),}$$

so the irreducibles stay exactly {one dimensionful unit,  $\pi$ }. *Re-typing:*  $v_{\text{geo}}$  is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

**Theorem G — the closure statement [O] (target)**

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure  $\mu_{\text{QG}}$ . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT  $R + R^2$  seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed* [verification/v28\_gravity\_fR.py].

## Certifiability and recommended order

| Building block                                | Machine                 | Hardness             |
|-----------------------------------------------|-------------------------|----------------------|
| $W_{\text{wall}}$ enumeration ( $C_U^{(1)}$ ) | Lean / Python / Wolfram | easy ( <i>done</i> ) |
| $D_4$ fixed-locus polynomial ( $C_U^{(2)}$ )  | Sage / Mathematica      | medium               |
| selector uniqueness ( $C_U^{(3)}$ )           | interval arithmetic     | medium               |
| SNF, determinants, spectra                    | Lean                    | easy–medium          |
| principal-symbol ellipticity (G1)             | xAct / Cadabra / Lean   | medium               |
| gap inequality (G5)                           | Lean                    | easy ( <i>done</i> ) |
| operator-norm bound (G5)                      | analysis + numerics     | hard                 |
| projective limit (G6)                         | not primarily machine   | very hard            |
| Ward identities (G7)                          | BV/BRST formal + checks | hard                 |

### Recommended order.

Selector-triangle pairings  $\rightarrow v_{\text{geo}}$  scale anchor  $\rightarrow G_{\text{net}}$  inclusion theorem

The flavor interface (historically  $U_{\text{wall}}$ ) is now reduced to the *selector-triangle pairings* (the dual normal pair  $(d, n)$  forcing  $R$  columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The  $v_{\text{geo}}$  scale anchor is the one dimensionful input (same nature as  $1/G$ ).  $G_{\text{net}}$  — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit.  $F_{\text{transfer}}$  (source $\rightarrow$ pole / relic / cosmology) is the remaining downstream interface.

$F_{\text{transfer}}$  is a **typed functor, not a bag of open topics**. It is standard physics fed TFPT *source* data,  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , with exactly four interfaces:

| Interface              | Map                                                                         | Status    | Source data $\rightarrow$ observable                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F_{\text{pole}}$      | source $\rightarrow$ pole masses                                            | [E] / [C] | Koide 53/54 = $a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([verification/v183_koide_f_corner_transfer.py])                                                      |
| $F_{\text{Boltzmann}}$ | CP source $\rightarrow \eta_B$                                              | [C]       | $\tilde{m}_1=m_3/A_\Lambda$ anchors the washout, but $M_1=M_R\varphi_0^4$ only relocates the free input ( $M_R$ is the seesaw scale, not a compiler power), so $\eta_B$ stays a <i>sharper scenario</i> , not a closure ([verification/v184_etaB_anchored_boltzmann.py]) |
| $F_{\text{relic}}$     | $(f_a, m_a, \theta_i, N_{\text{DW}}) \rightarrow \Omega_a$                  | [C]       | the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([verification/v185_axion_relic_solver.py])                                                             |
| $F_{\text{QCD}}$       | $(\alpha_s, b_3, \Lambda_{\text{QCD}}, \text{lattice}) \rightarrow m_p/m_e$ | [O]       | a QCD matching contract ( $b_3=-7$ a carrier output), <i>not</i> a compiler number — the rejected 1836 near-formula is the discipline working                                                                                                                            |

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([verification/v187\_ftransfer\_laws.py]). Exact *algebraic* sub-parts (the Koide factor,  $b_3=-7$ ) may be [E]; the *physical* prediction never is.

**The functor contract CONTRACT.F.01, four axioms** ([verification/v213\_ftransfer\_functor.py]). Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: (1)  **$\mu_4$ -deck equivariance** — the discrete kernels are carrier/deck invariants  $\{|\mu_4|, N_{\text{fam}}, g_{\text{car}}, A_\Lambda\}$ , and the Koide multiplier  $\lambda_2=(2/3)^6=64/729$  is the  $\mu_4$ -deck transfer eigenvalue (v82/v54); (2) **Plücker preservation** — the dimensionless readouts are Plücker readings ( $53=a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F)=(1, 22, 30)$ ;  $\|\text{Pl}(K)\|_1=11$ ); (3) **positivity/stochasticity** —  $\text{spec } T=\{1, (2/3)^6, (1/3)^6\}$  is positive (RP/OS),

$\kappa_f \in (0, 1)$ ; **(4) external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite- $T$   $\chi(T)$  ODE, the lattice  $C_N$ ) is named and external, never a compiler number. So `CONTRACT.F.01` is the third research contract alongside  $(U_{\text{wall}})$  and  $(G_{\text{metric}})$ : a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure.

**The discrete→dynamic lens: `F_transfer` is the readout end of the one principle** (`[verification/v303_ftransfer_dynamics.py]`, `FR.DYNAMICS.01`). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update  $m \leftarrow (A+2I)/4m \rightarrow$  the  $E_8$  marks, gap  $6 \ln(3/2)$ ). All four `F_transfer` instances share that *same shape*: **[E] `F_pole`** (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock subleading eigenvalue, so  $|F'| < 1 \Rightarrow$  a unique attractor at the *seam* rate (`v82`); **[C] `F_Boltzmann`** ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0, 1)$  toward the unique balance, an H-theorem); **[C] `F_relic`** (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); **[E] `F_QCD`** ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3=-7$ , whose Gaussian UV fixed point is the attractor (asymptotic freedom,  $\Lambda_{\text{QCD}}$  generated). *Honest boundary*: only `F_pole` has the seam rate  $(2/3)^6$  exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So `F_transfer` is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory’s simplicity is preserved precisely because the external rates are honestly fenced (`v187`), not because they reduce to the seam. (Even `F_pole`’s flow time is non-integer, `v93`: the relaxation is the *continuous* semigroup  $e^{t \log F}$  generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.) So the visible residual is one geometric postulate (`QGE0.SYM.01`), one unit gauge ( $v_{\text{geo}}$ ), and this one typed transfer functor — everything else is out-of-sample data.

$F_{\text{transfer}}$ : one shape, four rates — a gapped relaxation to a UNIQUE attractor

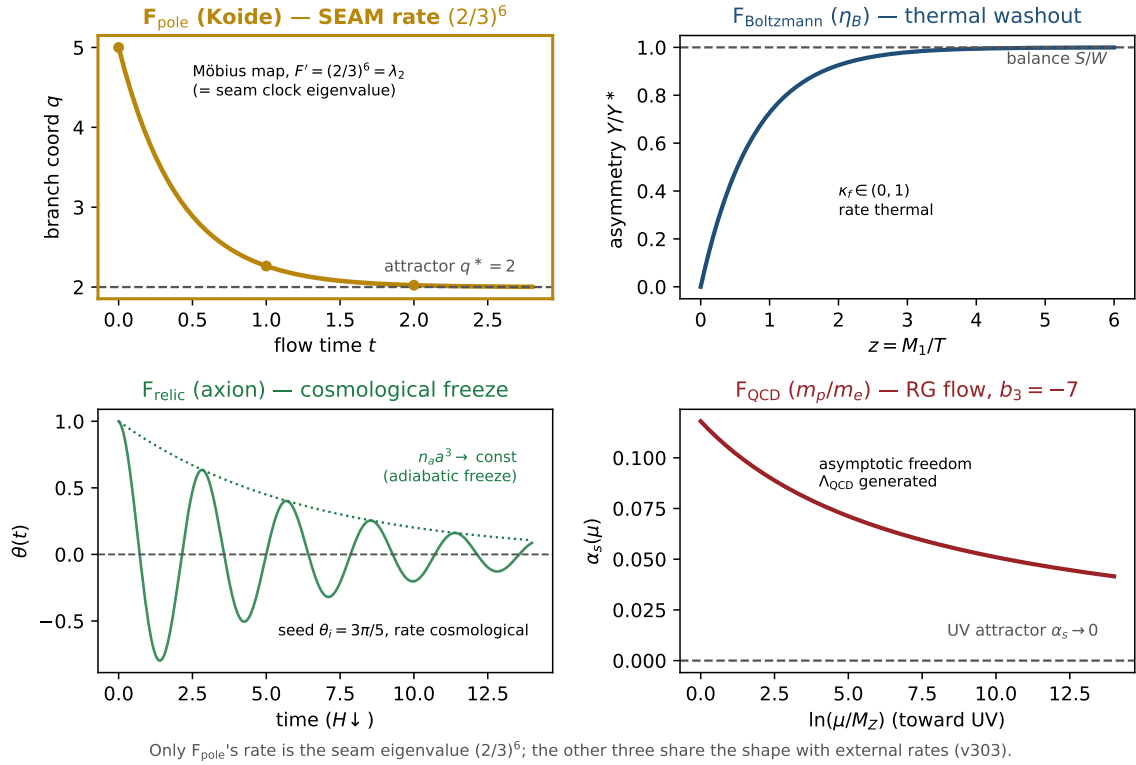


Figure 4: **The discrete→dynamic lens on  $F_{\text{transfer}}$**  ([verification/v303\_ftransfer\_dynamics.py]). All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update  $m \leftarrow (A+2I)/4m \rightarrow$  the  $E_8$  marks. *Only*  $F_{\text{pole}}$  (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is  $F'=(2/3)^6=\lambda_2$ , the seam-clock subleading eigenvalue.  $F_{\text{Boltzmann}}$  ( $\eta_B$ , thermal washout),  $F_{\text{relic}}$  (axion, cosmological freeze) and  $F_{\text{QCD}}$  ( $m_p/m_e$ , the RG flow to the Gaussian UV fixed point with the carrier  $b_3 = -7$ ) share the *shape* with *external* rates. So  **$F_{\text{transfer}}$**  is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.